

Confidence-Guided Selective LLM Re-Evaluation for Emotion Classification in Social Media

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ABSTRACT Confidence-guided routing enables selective LLM re-evaluation of difficult inputs while retaining the predictions of an efficient first-stage classifier for most posts. We develop and evaluate a two-phase pipeline for three-class, non-clinical proxy emotion classification in social media. A validation-selected DistilBERT model supplies calibrated predictions, and low-confidence posts undergo original-text re-evaluation. We compare Direct, Chain-of-Thought (CoT), and task-adapted SELF-DISCOVER with Llama 2 and Llama 3. On 12,000 Reddit posts, Phase 1 achieves 96.9167% accuracy and routes only 218 posts (1.82%); Llama 3 SELF-DISCOVER raises accuracy to 97.2000%, correcting 72 errors and introducing 38. On 300 synthetic Mixed Emotion examples, it increases accuracy from 83.6667% to 92.6667%, with 28 corrections and one introduced error. A separate 9,000-post same-source evaluation fixes the final classifier, prompts, and cached plan and routes 111 posts (1.23%). SELF-DISCOVER improves accuracy from 97.5778% to 97.7667%, yielding 17 net corrections. Its gains over the Phase 1 classifier are statistically significant on all three evaluation sets after the stated Holm adjustments; the additional evaluation gives $p = 0.0412$. Llama 3 SELF-DISCOVER also achieves the highest observed accuracy among the evaluated configurations on each set. These findings support confidence-guided selective re-evaluation as a means of improving classifier performance while limiting the number of posts sent to an LLM, and show the importance of model and prompting-protocol choice for balancing corrections against introduced errors.

INDEX TERMS Depression-risk-related emotion classification, social media text analysis, large language models, confidence-guided routing, Chain-of-Thought prompting, SELF-DISCOVER, mental health NLP, reasoning traceability

I. INTRODUCTION

The World Health Organization estimates that approximately 332 million people live with depression [1] and identifies it as a leading cause of ill health and disability [2]. Depression and anxiety together account for an estimated US\$1 trillion in annual productivity losses [3]; their prevalence also increased during the first year of the COVID-19 pandemic [4]. This burden motivates research on emotional signals in everyday language, including computational analysis of non-clinical social-media text.

Validated questionnaires such as the Patient Health Questionnaire-9 (PHQ-9) support depression screening [5], while accessible, person-centred care remains a health-system priority [6]. Social-media analysis offers a complementary research setting for studying linguistic and emotional patterns [7]. We focus on post-level proxy emotion labels rather than clinical diagnosis.

Domain-adapted models such as MentalBERT demonstrate the value of target-domain pretraining for mental health

text classification [8]. Nevertheless, emotion-fusion research identifies dataset quality and interpretability as persistent challenges [9]. User-level prediction can additionally benefit from temporal and inter-post information [10]. Research on clinical explainability emphasizes that explanations should match the intended setting and information needs [11]. In our post-level setting, textual rationales accompany label decisions, while predictive value is assessed through observed corrections and introduced errors.

Existing research establishes the value of transformer classification, confidence calibration, selective prediction, and LLM-generated explanations, but these components answer different questions and are not interchangeable. A calibrated classifier may identify difficult inputs without correcting them, while an LLM may generate a plausible rationale without improving the underlying decision. This leaves a practical evaluation gap: confidence-based case selection and reasoning-based correction should be connected in one fixed pipeline and

assessed separately at the classifier, router, and re-evaluator levels.

To address this gap, we propose a confidence-guided two-phase framework for depression-risk-related proxy emotion classification that combines transformer-based classification with selective, rationale-supported LLM re-evaluation. The completed Phase 1 implementation uses DistilBERT to produce initial predictions and calibrated maximum-softmax-probability confidence scores. A dedicated calibration split is held out from model selection and used for temperature scaling, calibration assessment, and routing-threshold selection. Samples whose calibrated confidence scores fall below the selected threshold are routed to Phase 2. The threshold maximizes Phase 1 coverage subject to a one-sided upper bound on accepted-set selective risk. In Phase 2, low-confidence samples are re-analyzed through model-specific structured reasoning processes [12], [13]. The resulting end-to-end evaluation reports not only final accuracy, but also whether routing concentrates errors and whether each re-evaluator corrects more routed errors than it introduces.

The evaluation addresses three research questions, with a separate same-source evaluation of the fixed final Llama 3 protocols. **RQ1a** assesses the operational classifier's predictive and calibration performance; **RQ1b** compares the candidate classifiers under a shared data and search budget. **RQ2** asks whether the fixed routing policy concentrates Phase 1 errors while retaining high coverage. **RQ3** asks whether re-evaluation produces positive net corrections on Reddit posts and on a controlled synthetic stress test. Separating these questions makes the contribution of case selection and subsequent correction measurable within the same pipeline.

II. STUDY CONTRIBUTIONS

The study makes four concrete contributions:

- It operationalizes selective LLM re-evaluation as an end-to-end pipeline in which an efficient classifier handles high-confidence inputs and reasoning models receive only calibration-defined low-confidence inputs. The Phase 1 comparison grounds model selection in predictive performance and measured inference throughput, latency, and memory on a common accelerator under a defined search budget.
- It replaces a heuristic confidence cutoff with a reproducible calibration protocol: temperature scaling, a pre-specified threshold grid, a one-sided selective-risk upper bound, and maximum-coverage selection among feasible thresholds. The selected policy is fixed before held-out and Phase 2 evaluation.
- It evaluates routing quality separately from re-evaluation quality. Coverage, selective risk, error capture, corrected errors, introduced errors, and net corrections reveal whether the router finds difficult inputs and whether the LLM actually improves them. The final comparison crosses two language models with three re-evaluation protocols on the same routed examples within each dataset.

- It develops a task-specific SELF-DISCOVER adaptation [13] combining an emotion-oriented module bank, a reusable model-generated plan, and fixed checks for emotional subject, temporal context, competing evidence, and alternative interpretations. The adaptation is evaluated on the primary Reddit test set, a controlled Mixed Emotion stress test excluded from training and threshold selection, and a separate 9,000-post evaluation with the final Llama 3 protocols fixed. Example-aligned predictions quantify corrected and introduced errors.

Together, these contributions connect efficient classifier selection, calibrated routing, and task-adapted LLM re-evaluation in a unified, auditable pipeline. Llama 3 SELF-DISCOVER yields positive net corrections and statistically significant accuracy gains over Phase 1 on all three evaluation sets under the stated Holm adjustments. Within-model comparisons further characterize how prompting protocols affect correction yield across datasets; detailed paired results are reported in the results section and Appendix E.

III. LITERATURE REVIEW

A. EARLY APPROACHES: MACHINE LEARNING FOR DEPRESSION DETECTION

Early approaches used linguistic and behavioral features with conventional classifiers. In a study of college-student essays, Rude et al. [14] found more negative-emotion words and more frequent use of "I" among currently depressed than never-depressed participants. This finding concerns a specific population and writing task, rather than a universal diagnostic marker. De Choudhury et al. [15] used linguistic style, engagement, and network features to predict depression-related outcomes from Twitter activity. Tsugawa et al. [16] evaluated Twitter activity features with an SVM against questionnaire-based depression scores. CLPsych 2015 subsequently provided shared user-level classification tasks based on self-reported diagnoses and matched controls [17].

Feature-based approaches also included Shen et al.'s multi-modal dictionary-learning model, which combined six groups of depression-related and behavioral features from Twitter [18]. Tadesse et al. [19] evaluated linguistic and topic-based features with several classifiers on 1,841 Reddit posts, comprising 1,293 depression-indicative and 548 comparison posts. These studies established useful predictive baselines without making their labels equivalent to clinical assessments. Wongkoblap et al. [20] identified data-collection bias, consent, dataset quality, and methodological choices as continuing limitations of social-media mental-health research.

B. DEEP LEARNING FOR CONTEXTUALIZED DEPRESSION DETECTION

Neural approaches learn text representations rather than relying exclusively on handcrafted features. Word embeddings provide dense representations that encode semantic relationships [21], [22]. LSTMs were designed to learn dependencies over extended sequences [23], while CNNs operating on word vectors were evaluated for sentence classification [24].

Orabi et al. (2018) [25] compared several deep learning models for user-level depression detection on Twitter. Using a TF-IDF linear SVM as a baseline, they showed that CNN-based models, particularly those using their optimized embeddings, outperformed the baseline classifier. In their experiments, CNN models also achieved better performance than the BiLSTM model. The authors further noted that supervised deep neural approaches have seen limited adoption in this domain due to the difficulty of obtaining sufficiently annotated training data [25].

Shared evaluations also changed the prediction setting. The eRisk 2018 depression task released users' posts in chronological chunks and evaluated early decisions [26]. This user-history setting differs from the independent post-level classification studied here; results across the two tasks are not directly comparable.

C. LARGE LANGUAGE MODELS FOR MENTAL HEALTH ANALYSIS

Transformer encoders and generative LLMs serve different roles in mental-health text analysis. Encoders provide contextual representations for supervised classification [8], [27], [28], whereas generative models can produce labels together with textual responses. Their use in the same application domain does not make predictive accuracy and explanation quality interchangeable outcomes.

Zhang et al. [27] studied transformer-based depression-risk detection using Sina Weibo posts. MentalBERT [8] instead emphasizes domain-specific pretraining, while ensemble approaches combine transformer classifiers [28]. These studies motivate contextual representations but do not determine whether a second-stage model should process every post or only a selected subset.

Shin et al. [29] evaluated GPT-3.5 and GPT-4 on semistructured emotional diaries, using questionnaire-based assessments as reference measures. Their study supports investigating diary text for depression-related screening, but its participants, reference measures, and input format differ from subreddit-derived post labels.

Wang et al. [30] used Reddit writings to predict questionnaire-related depression levels and generate explanations for individual questionnaire items. Lan et al.'s DORIS framework [31] combines LLM-derived symptom annotations and temporal mood-course features with a gradient-boosting-tree classifier and generated justifications. These studies illustrate distinct ways of integrating prediction and explanation. Omar and Levkovich's review [32] identifies predictive potential across encoder and generative models while emphasizing validation, privacy, and ethical safeguards.

These approaches retain the construct-validity, sampling, and platform-dependence concerns documented in reviews [33], [34], [35]. For the present task, a generated explanation is therefore an inspectable output, not evidence that the proxy label or the prediction is clinically valid.

D. CONFIDENCE CALIBRATION, SELECTIVE PREDICTION, AND CASCADED RE-EVALUATION

Calibration addresses the correspondence between predicted confidence and observed correctness [36]. Selective classification addresses a related but distinct problem: trading coverage for accepted-set risk by withholding an immediate prediction on selected inputs, a strategy termed abstention [37]. Abstention does not itself calibrate probabilities or correct the rejected predictions. This distinction motivates separate evaluation of confidence quality, case selection, and downstream correction.

Here, rejected inputs are sent to an LLM rather than left unclassified. Temperature scaling [36] and selective prediction [37] motivate the routing design; CoT [12] and SELF-DISCOVER [13] motivate the adapted re-evaluation protocols described below. The empirical question is whether recombining their outputs improves the original classifier after newly introduced errors are counted.

E. DATASET CONSIDERATIONS IN SOCIAL MEDIA-BASED MENTAL HEALTH TEXT ANALYSIS

Progress in AI-based depression-related text analysis has been strongly shaped by the availability and quality of datasets. Early studies often relied on relatively small, task-specific datasets collected from social media platforms where users publicly disclosed a diagnosis of depression. A representative example is the CLPsych 2015 Shared Task dataset, which comprised Twitter users who self-reported depression and demographically matched control users. This dataset served as an early benchmark for depression-related classification from social media text [17].

As research in this area expanded, social media corpora became increasingly important. For example, Tadesse et al. analyzed Reddit posts using natural language processing and machine learning techniques to identify depression-related content [19]. Twitter was reported as the most frequently studied platform for social media-based depression-related analysis, while Reddit and Facebook were also used in this line of research [33].

Despite their usefulness, these datasets have important limitations. Many rely on self-disclosed diagnoses and indirectly constructed labels, raising concerns about construct validity and the operationalization of mental health status in social media-based research. Chancellor and De Choudhury identified inconsistencies in label operationalization, the lack of standardized evaluation practices, and the need for greater transparency in data curation and validation procedures in social media-based mental health research [34]. More recent review evidence similarly emphasizes that social-media depression detection studies remain vulnerable to sampling bias, platform bias, language and demographic imbalance, inconsistent preprocessing, and incomplete reporting of model-development procedures [35]. These concerns motivate the present study's explicit treatment of subreddit-derived labels as proxy emotion labels rather than clinical labels, as well as

its reporting of data construction, filtering, calibration, and threshold-selection procedures.

Privacy and ethical concerns also remain central in mental health-related text analysis using social media data [38], [39]. Because such data may contain sensitive personal information, responsible data handling practices and clearly defined research protocols are essential for ethical and rigorous research [39].

IV. DATASET AND PREPROCESSING

The primary Reddit corpus supplies model development and evaluation data. A separate synthetic dataset tests controlled affective scenarios and Neutral controls. Their construction and uses are described below.

A. DATASET CONSTRUCTION

The primary dataset was constructed from Reddit posts collected from subreddit communities used as class proxies. Depression and AnxietyDepression supplied the Depression group; technology, datascience, AskScienceDiscussion, and webdev supplied the Neutral group; and Positivity, happy, MadeMeSmile, UnexpectedlyWholesome, and CongratsLikeImFive supplied the Happy group. After cleaning, polarity filtering, and class balancing, the preprocessing artifact contained 133,660 rows for three-class emotion classification. Each row corresponds to a sampled Reddit post, and the post title and body were concatenated before cleaning. Because community membership supplies a proxy label rather than a clinical annotation, community-specific lexical cues and sampling choices are potential sources of construct bias.

The Depression class was intended to represent posts containing depression-related emotional expressions or distress-related language. The Neutral class represented posts with a relatively neutral affective tone, whereas the Happy class represented posts containing positive emotional expressions. These labels were used as proxy labels for depression-risk-related emotion classification and should not be interpreted as formal clinical diagnostic labels.

B. DATA FEATURES

For each Reddit post, the text fields and metadata needed for preprocessing, linkage, and evaluation were retained during dataset construction. The Phase 1 modeling input was the cleaned title–body text (`Title_with_selftext_cleaned`), which was generated by concatenating the post title and body text. The retained original title and selftext were used only when a routed post entered Phase 2. Additional fields, including subreddit source, author-related metadata, adult-content flag, timestamp, polarity score, and class label, were used only to support preprocessing, sampling, linkage, and dataset organization. A complete description of the retained variables is provided in Appendix Table A1.

Class labels were numerically encoded for model training as Depression = 0, Neutral = 1, and Happy = 2. This encoding was used only for computational convenience and does not imply an ordinal relationship among the classes.

C. TEXT CLEANING AND NORMALIZATION

A standardized preprocessing pipeline was applied to reduce noise and improve comparability across posts. The concatenated title and selftext fields were first converted to lowercase and tokenized. Non-alphabetic content, URLs, numbers, special characters, and common stop words were removed. Lemmatization was then applied to reduce inflected word forms to their base forms. For example, “Sad” was converted to “sad,” and “running” was lemmatized to “run.”

In addition, a Reddit-specific stop-word list filtered platform-specific filler expressions. The cleaned field, `Title_with_selftext_cleaned`, was used for Phase 1 training, calibration, and prediction. Cleaning preserves the order of retained tokens but can remove negation, punctuation, and discourse markers. For routed Reddit posts, Phase 2 used the linked original title and selftext, not the Phase 1-cleaned input. It decoded HTML, replaced URLs and direct usernames, and normalized whitespace while preserving wording, capitalization, punctuation, negation, sentence order, and sentence boundaries.

D. SENTIMENT-AWARE DATA SAMPLING

To support label consistency, a sentiment-aware filtering step was applied after text cleaning. The implementation passes the complete cleaned title–body string to `TextBlob` and reads its document-level polarity value in the range $[-1, 1]$, as documented by `TextBlob` [40]. `Pattern` provides related NLP and sentiment-analysis functionality [41]; the expression below specifies the `TextBlob` call used here:

$$P_{post} = \text{TextBlobPolarity}(\text{Clean}(\text{title} \oplus \text{body})), \quad (1)$$

where $\oplus$ denotes title–body concatenation and `Clean(·)` denotes the cleaning pipeline described above. Values near -1 , 0 , and $+1$ indicate increasingly negative, neutral, and positive polarity, respectively. This expression matches the document-level software call used by the implementation; the study code does not compute a separate `TextBlob` score for every token and then average those scores.

Class-specific polarity thresholds were then applied to retain posts whose overall sentiment broadly aligned with the intended emotional label. Specifically, posts in the Depression class were retained if their polarity scores fell within $[-1, 0.2]$, Neutral posts were retained within $[-0.3, 0.3]$, and Happy posts were retained within $[-0.2, 1]$. Posts whose polarity scores were substantially inconsistent with their intended labels were excluded during dataset construction. This procedure was intended to reduce obvious sentiment-label mismatch while preserving the three-class emotion classification setting.

E. MIXED EMOTION DATASET

In addition to the primary Reddit dataset, we constructed a controlled synthetic Mixed Emotion stress-test dataset to examine model behavior under emotionally ambiguous conditions. The final dataset contains 300 controlled examples, balanced

across the three proxy emotion labels, with 100 Depression, 100 Neutral, and 100 Happy examples. This dataset was not used for Phase 1 model training, hyperparameter tuning, temperature scaling, or routing-threshold selection. Instead, it was designed as a targeted stress-test set for cases in which positive, neutral, and depression-related cues co-occur or shift across the text. In the evaluation workflow, the temperature and routing threshold selected on the Reddit calibration split are fixed and then applied directly to the Mixed Emotion Dataset; thresholds are not re-selected on the synthetic stress-test data.

The current dataset includes seven controlled scenario groups: blended-emotion co-occurrence (40), positive-to-distress shift (40), distress-to-recovery shift (40), neutral framing with subtle affect (40), conflicting cues with a dominant trajectory (40), clear technical or informational Neutral examples (75), and Neutral examples with mild factual or procedural ambiguity (25). Each example was assigned a target label according to the dominant overall emotional trajectory rather than isolated sentiment-bearing phrases. The two Neutral groups preserve the proxy-label definition while avoiding a stress set that is artificially dominated by either completely trivial or clinically suggestive Neutral content. This design tests whether confidence-guided routing and reasoning-based re-evaluation can address cases containing competing or shifting emotional evidence.

Synthetic examples were generated using controlled prompting and checked against inclusion and exclusion criteria. Kang et al. [42] used LLM-generated clinical-interview summaries for depression-prediction training augmentation. Here, synthetic posts provide controlled stress-test evaluation, distinct from training augmentation or clinical validation. Inclusion required consistency with the assigned scenario and target label: mixed or shifting cues for the affective scenarios, and low-affect content for the Neutral controls. Exclusions covered explicit clinical diagnosis claims, treatment recommendations, crisis language, identifying information, and off-scenario content. Appendix Table F1 records the generation protocol. These checks establish adherence to the dataset design criteria. Independent expert annotation and naturally occurring mixed-emotion evaluation remain outside the scope of this dataset.

F. DATA PARTITIONING AND MODEL INPUTS

After preprocessing and label encoding, the primary Reddit dataset is divided into training, model-validation, threshold-calibration, and held-out test sets by independently shuffling and partitioning each class. The final protocol samples exactly 40,000 examples per class (120,000 in total) and assigns 28,000 examples per class to training and 4,000 examples per class to each of validation, calibration, and held-out testing. Thus, the final split sizes are 84,000 training, 12,000 validation, 12,000 calibration, and 12,000 test examples. This per-class construction avoids one-example rounding differences from successive proportion-based split operations.

The training partition supplies supervised parameter updates; model validation supplies hyperparameter, epoch, and checkpoint selection. The calibration partition supplies temper-

ature fitting and threshold selection. Test labels are excluded from these Phase 1 operations. The routed Reddit test cases also informed Phase 2 prompt development; within-set prompt-configuration rankings are therefore interpreted as exploratory. Each classifier uses its own tokenizer on the cleaned title-body field.

Only the cleaned textual content was used as the direct Phase 1 classifier input. For routed Reddit examples, Phase 2 received the linked original title and selftext after minimal privacy-oriented sanitization. The remaining metadata, such as subreddit source, author-related metadata, timestamp, and adult-content flag, was not used as a direct predictive feature in either phase.

G. ADDITIONAL SAME-SOURCE EVALUATION

A separate balanced evaluation set comprises 9,000 posts (3,000 per class) sampled from records in the same source corpus that were not included in the original 120,000-post sample. Selection excludes normalized exact-text matches to the original sample and duplicate normalized texts within the candidate pool using Unicode NFKC normalization, case folding, and whitespace normalization. Sampling uses a fixed random seed. These procedures remove normalized exact-text overlap under the specified rule; the set remains same-source and is not guaranteed to be disjoint by author or time.

Before evaluating this set, we fixed the final seed-42 DistilBERT model trained under the selected hyperparameters, the calibration temperature, the routing threshold, the three Llama 3 prompting protocols, and the previously generated SELF-DISCOVER plan. No training, recalibration, threshold selection, SELF-DISCOVER plan generation, or prompt revision used these 9,000 posts. Direct, CoT, and SELF-DISCOVER evaluated the same routed cases, and all end-to-end metrics were computed over the full 9,000-post set.

H. ETHICAL CONSIDERATIONS

Because this study analyzes depression-risk-related emotional signals from social media text, several ethical considerations are relevant to dataset construction and interpretation.

First, the dataset labels are proxy labels derived from social media context and sentiment-aware filtering, not diagnoses assigned by licensed mental health professionals. Therefore, the proposed framework should be interpreted as a research-oriented emotion classification system rather than a clinical diagnostic or screening tool.

Second, Reddit posts may contain sensitive personal expressions related to emotional distress or mental health. To reduce privacy risks, personally identifying information was not used as a modeling feature. Before routed original text was passed to Phase 2, URLs were replaced by [URL] and direct Reddit username patterns were replaced by [USER]; sentence order, punctuation, capitalization, and negation were otherwise preserved. Author-related metadata was used only for dataset organization and was not used for prediction. These measures are limited sanitization, not a guarantee of anonymity: retained verbatim passages may remain searchable. The analysis fo-

cused on aggregate model development and evaluation rather than individual-level identification or profiling.

Third, the Mixed Emotion Dataset was generated through controlled prompting with GPT-5 and retained as a versioned evaluation artifact. Because synthetic examples may reflect generation-model biases or stylistic patterns, this dataset was used only for stress-test evaluation and not for primary model training. Future work should validate the framework with naturally occurring and expert-reviewed datasets before considering any real-world decision-support use.

Finally, the framework was not designed to identify, monitor, or intervene with individual users. Any deployment beyond research use would require institutional review, domain-expert oversight, bias and fairness assessment, and compliance with relevant data governance requirements.

V. METHODOLOGY

A. UNIFIED CLASSIFIER-TO-RE-EVALUATION PIPELINE

The final DistilBERT configuration is selected through the Phase 1 validation-based search. A final training run under that configuration supplies the predictions, logits, calibration temperature, and routed IDs used throughout the downstream experiment. No separately configured operational classifier is substituted after the classifier comparison. Three-seed classifier summaries describe training variability; downstream results describe this single linked final run, not an average of three reasoning pipelines.

At inference, accepted predictions retain their Phase 1 labels. Every routed post is independently evaluated under each of six configurations: Llama 2-7B-Chat or Llama 3-8B-Instruct, each with Direct, CoT, or final SELF-DISCOVER prompting. These are separate alternatives, not an ensemble or a dataset-specific selection rule. All configurations within a dataset use the same Phase 1 outputs and routed IDs. Figure 1 summarizes this shared pipeline; Figure 2 details the SELF-DISCOVER procedure.

B. PHASE 1: INITIAL EMOTION CLASSIFICATION

The Phase 1 comparison includes DistilBERT [43], Mistral 7B [46], and Llama 2 7B [44] under common data partitions and a four-trial search budget. DistilBERT is fully fine-tuned, whereas the 7B backbones use frozen representations with trainable linear heads. The comparison therefore evaluates practical classifier configurations under shared data partitions and a bounded search budget. Each selected configuration is evaluated with seeds 42, 43, and 44. Appendix Table A1c gives the selected settings; the final DistilBERT run under the selected configuration supplies downstream routing scores.

C. DISTILBERT

DistilBERT [43] is a compact bidirectional encoder obtained by distilling BERT into a six-layer student. Its original pre-training combines language-modeling, teacher-distribution matching, and representation-alignment objectives, producing a model with about 40% fewer parameters than BERT-base and a reported inference speedup of about 60% while retaining

much of the teacher’s language-understanding performance. Those properties make it an appropriate candidate for the high-volume first stage of a selective pipeline, where every input must be classified before only a small subset is escalated. In this study, the released `distilbert-base-uncased` checkpoint is fully fine-tuned with supervised three-class cross-entropy; the historical distillation process is not repeated. Its logits are subsequently temperature-calibrated, and only this operational checkpoint supplies the confidence scores used for routing.

D. MISTRAL

Mistral 7B [46] is a decoder-only transformer with grouped-query attention (GQA) and sliding-window attention (SWA). GQA reduces key-value memory requirements, while SWA restricts attention to a local window. Here, Mistral provides a larger pretrained representation for three-class classification. Its backbone remains frozen, and a bias-free linear head maps the 4,096-dimensional representation to three class logits. Only the head’s 12,288 parameters are trained; no adapter is added.

E. LLAMA 2

Llama 2 [44] is a family of decoder-only autoregressive transformers. Its 7B base checkpoint supplies a second larger representation family for Phase 1 and is distinct from the chat-tuned checkpoint used in Phase 2. Classification uses the same frozen-backbone, bias-free 12,288-parameter linear-head design as Mistral. The shared partitions and label mapping permit predictive comparison, while the differing adaptation scope precludes attributing differences to model size alone.

All three classifiers output class logits. Post-hoc calibration and threshold selection are performed only for the operational DistilBERT checkpoint; no calibrated routing result is claimed for the comparison models.

F. CALIBRATED CONFIDENCE-BASED ROUTING AND RISK-COVERAGE THRESHOLD SELECTION

The fixed operational DistilBERT classifier supplies the logits used for calibration and routing. For an input text x_i , it outputs logits z_i over K proxy emotion classes. The uncalibrated class probability for class k is computed with the softmax function:

$$p_{i,k} = \frac{\exp(z_{i,k})}{\sum_{j=1}^K \exp(z_{i,j})}. \quad (2)$$

The Phase 1 predicted label is defined as:

$$\hat{y}_i = \arg \max_k p_{i,k}. \quad (3)$$

Because raw maximum softmax probability may be poorly calibrated [36], a dedicated calibration split D_{cal} is used to fit a scalar temperature parameter T . Temperature-scaled probabilities are computed as:

$$q_{i,k}(T) = \frac{\exp(z_{i,k}/T)}{\sum_{j=1}^K \exp(z_{i,j}/T)}. \quad (4)$$

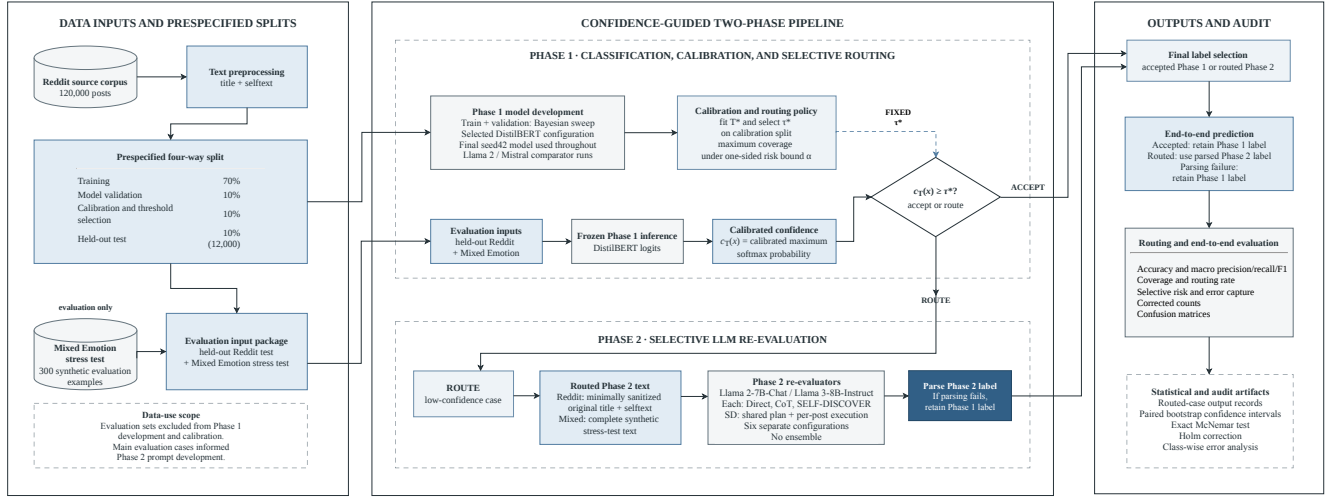


FIGURE 1. Confidence-guided two-phase pipeline. Accepted predictions retain the Phase 1 label, while routed posts enter re-evaluation. Llama 2 and Llama 3 are each evaluated with Direct, CoT, and SELF-DISCOVER on the same routed posts. Parsed labels replace the initial predictions; parsing failures retain Phase 1 labels.

The temperature is selected by minimizing negative log-likelihood on D_{cal} :

$$T^* = \arg \min_{T>0} \frac{1}{|D_{cal}|} \sum_{i \in D_{cal}} -\log q_{i,y_i}(T). \quad (5)$$

NLL evaluates the probability assigned to the true class and penalizes confident mistakes strongly. Two complementary diagnostics are also reported. The multiclass Brier score measures the squared distance between the complete probability vector and the one-hot target vector:

$$\text{Brier}(T) = \frac{1}{|D_{cal}|} \sum_{i \in D_{cal}} \sum_{k=1}^K (q_{i,k}(T) - \mathbf{1}[y_i = k])^2. \quad (6)$$

Expected calibration error (ECE) partitions predictions into confidence bins B_m and compares mean confidence with empirical accuracy within each bin:

$$\text{ECE}(T) = \sum_{m=1}^M \frac{|B_m|}{|D_{cal}|} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (7)$$

NLL, Brier score, and ECE are complementary rather than interchangeable selection constraints. NLL is the optimization objective for T^* ; Brier score checks the quality of all class probabilities; and ECE checks whether stated confidence agrees with observed correctness at the group level. Adaptive ECE repeats the last comparison with approximately equal-frequency bins to reduce sensitivity to sparse fixed-width bins. Lower values are better for all four diagnostics.

The primary routing confidence score is then the temperature-scaled maximum softmax probability:

$$c_i = \max_k q_{i,k}(T^*). \quad (8)$$

For a candidate routing threshold τ , the accepted set $A(\tau)$ and routed set $R(\tau)$ are defined as:

$$A(\tau) = \{i : c_i \geq \tau\}, \quad (9)$$

$$R(\tau) = \{i : c_i < \tau\}. \quad (10)$$

Coverage and routing rate are computed as:

$$\text{Coverage}(\tau) = \frac{|A(\tau)|}{N}, \quad (11)$$

$$\text{RoutingRate}(\tau) = \frac{|R(\tau)|}{N} \quad (12)$$

$$= 1 - \text{Coverage}(\tau). \quad (13)$$

Selective risk is computed only on accepted Phase 1 predictions:

$$\text{Risk}(\tau) = \frac{1}{|A(\tau)|} \sum_{i \in A(\tau)} \mathbf{1}[\hat{y}_i \neq y_i]. \quad (14)$$

To make the routing rule more conservative on finite calibration samples, the implementation also computes a one-sided binomial upper confidence bound $\bar{R}_\delta(\tau)$ for the accepted-set error rate using confidence level $1 - \delta$. The final threshold is selected on D_{cal} by maximizing coverage while satisfying the target selective-risk constraint α :

Let $e(\tau) = \sum_{i \in A(\tau)} \mathbf{1}[\hat{y}_i \neq y_i]$ denote the number of accepted-set errors and $n(\tau) = |A(\tau)|$ denote the number of accepted examples. For $e(\tau) < n(\tau)$, the one-sided Clopper-Pearson upper bound used by the implementation is:

$$\bar{R}_\delta(\tau) = \text{Beta}^{-1}(1 - \delta; e(\tau) + 1, n(\tau) - e(\tau)), \quad (15)$$

where $\text{Beta}^{-1}(\cdot; a, b)$ is the inverse cumulative distribution function of a $\text{Beta}(a, b)$ distribution. When every accepted prediction is incorrect, the bound is set to 1; empty accepted

sets are ineligible. The tail probability δ specifies the nominal confidence level, whereas α specifies the acceptable risk budget. This is a calibration-set selection criterion. Because temperature fitting and threshold search reuse the calibration data, pointwise Clopper–Pearson bounds do not establish a simultaneous post-selection guarantee for the resulting policy. Nor does the criterion guarantee risk control under distribution shift.

$$\tau^* = \arg \max_{\tau \in \mathcal{T}} \text{Coverage}(\tau) \quad (16)$$

$$\text{s.t. } \bar{R}_\delta(\tau) \leq \alpha, \quad (17)$$

$$|A(\tau)| \geq m_{\min}. \quad (18)$$

Here, $m_{\min}$ is a minimum accepted-sample requirement used to avoid selecting thresholds supported by too few calibration examples. The implementation sets $m_{\min} = \lceil 0.10|D_{\text{cal}}| \rceil = 1,200$. The selected $\tau = 0.70$ accepted 11,807 calibration examples, so this safeguard was satisfied but did not determine the selected operating point. In the final DistilBERT operating policy, candidates are swept from 0.70 to 1.00 in increments of 0.01. This lower bound was chosen before held-out testing to avoid treating a weak three-class majority probability near 0.50 as sufficiently reliable for direct acceptance. Unless otherwise stated, the primary operating point uses temperature-scaled maximum softmax probability, a target selective risk of $\alpha = 0.05$, a one-sided risk confidence level of $1 - \delta = 0.95$, the acceptance rule $c_i \geq \tau$, and the routing rule $c_i < \tau$. The risk constraint is a feasibility condition: it excludes candidate thresholds whose accepted-set upper risk exceeds α . Among the feasible candidates in the prespecified 0.70–1.00 grid, the candidate with the greatest Phase 1 coverage is selected. Thus, the completed DistilBERT value $\tau^* = 0.70$ reflects both the prespecified conservative operating range and calibration-set risk–coverage evaluation; it was not optimized using held-out test results or Phase 2 outcomes. The 7B models serve as Phase 1 predictive comparators only; they do not define alternative routing policies in the reported end-to-end experiment.

The threshold-selection output records accepted accuracy, routed Phase 1 errors, error capture, routing precision, and proxy-Depression false-negative risk among accepted predictions. Additional confidence diagnostics include expected calibration error (ECE), adaptive ECE, Brier score, and negative log-likelihood before and after temperature scaling. Supporting analyses compare raw MSP, temperature-scaled MSP, entropy-based certainty, and probability margin; estimate bootstrap confidence intervals for selective metrics; assess threshold stability through calibration-set resampling; and extract high-confidence accepted errors for qualitative auditing.

Algorithm 3 formalizes the evaluation boundary between routed-subset behavior and full-set performance. In particular, a Phase 2 model is not credited merely for correcting selected errors: any newly introduced errors are deducted, and the final accuracy is recomputed over the complete evaluation

Algorithm 1 Calibrated confidence-guided routing

```

1: Input:  $D_{\text{val}}, D_{\text{cal}}$ , classifier  $f_\theta$ 
2: Input:  $\alpha, 1 - \delta$ , grid  $\mathcal{T}$ , minimum count  $m_{\min}$ 
3: Output: temperature  $T^*$  and threshold  $\tau^*$ 
4: Select best checkpoint using  $D_{\text{val}}$ 
5: Fit  $T^*$  on  $D_{\text{cal}}$  by minimizing NLL
6: Compute calibrated confidences  $c_i$  on  $D_{\text{cal}}$ 
7: Sweep the prespecified calibrated-confidence candidate grid
8: for each  $\tau \in \mathcal{T}$  do
9:   Form  $A(\tau)$  and  $R(\tau)$  from  $c_i$ 
10:  Compute coverage, routing rate, and risk bound
11: end for
12: Retain candidates with  $\bar{R}_\delta(\tau) \leq \alpha$  and  $|A(\tau)| \geq m_{\min}$ 
13: If none remain, flag infeasibility; a fallback is not risk-feasible
14: Choose a retained  $\tau$  with maximum coverage
15: Fix  $T^*$  and  $\tau^*$  before held-out testing
  
```

Algorithm 2 Two-phase inference with fixed calibrated routing policy

```

1: Input: post record  $x$ , classifier  $f_\theta$ , fixed  $T^*$  and  $\tau^*$ 
2: Input: Phase 2 configuration  $g$  (model and prompting protocol)
3: Obtain the Phase 1 input  $x^{(1)}$  from record  $x$ 
4: Compute  $\mathbf{z} = f_\theta(x^{(1)})$  and probabilities  $q_k(T^*)$ 
5: Set  $\hat{y}^{(1)} = \arg \max_k q_k(T^*)$  and  $c = \max_k q_k(T^*)$ 
6: if  $c \geq \tau^*$  then
7:   return  $\hat{y}^{(1)}$  as the accepted Phase 1 final label
8: else
9:   Retrieve the minimally sanitized original text  $x^{(2)}$ 
10:  Generate a label with  $g$  using its prescribed inputs
11:  Parse exactly one label from {Depression, Neutral, Happy}
12:  If parsing fails, retain  $\hat{y}^{(1)}$  and record failure
13:  return the parsed label or recorded fallback
14: end if
  
```

set after accepted Phase 1 labels and routed Phase 2 labels are recombined.

The selected calibration point satisfied the upper-bound criterion, so the export notebook’s infeasibility fallback was not invoked. The stored calibration, threshold, and environment artifacts record the policy values and model and seed identifiers.

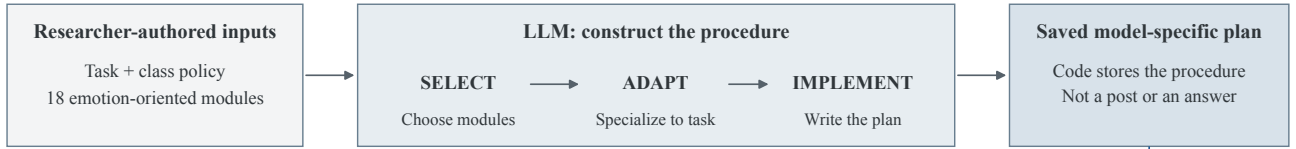
G. RELATIONSHIP TO MIXED-EMOTION EVALUATION

The routing mechanism is entirely confidence-based during inference. It does not directly detect blended emotion or sentiment shift. Instead, the Mixed Emotion Dataset serves a complementary evaluation role by stress-testing whether the Phase 1 classifier and the Phase 2 reasoning stage behave more reliably on examples where the one-label-per-input assumption is more difficult. This separation keeps the routing rule reproducible while allowing the evaluation to probe emotionally complex cases.

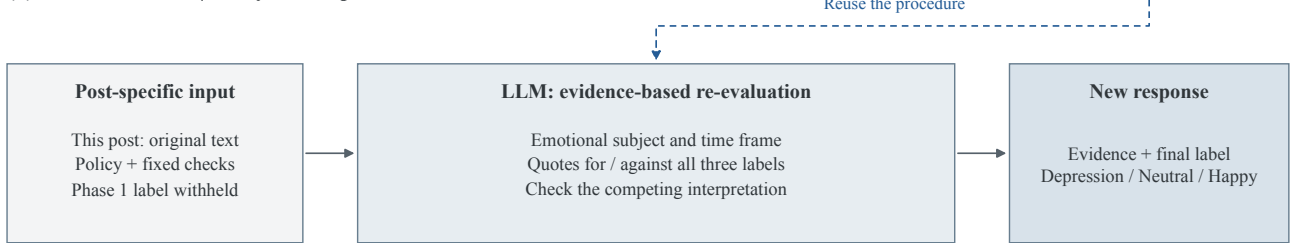
H. PHASE 2: FACTORIAL MODEL AND PROTOCOL COMPARISON

Direct prompting requests one final label without a rationale and includes the Phase 1 prediction. CoT first requests an independent assessment, subsequently discloses the Phase 1 label for comparison, and finally requests a terminal label. The implemented CoT dialogue contains five generation calls, including the initial acknowledgment and text-response turns. Final SELF-DISCOVER withholds the Phase 1 label

(a) Plan construction | Before post classification



(b) Fresh execution | Every routed post



Discovery uses no evaluation post or reference label. A saved plan bypasses all three discovery calls.

Each LLM retains its own plan across both datasets; every routed post receives a new execution response.

FIGURE 2. Task-specific SELF-DISCOVER design within selective re-evaluation. (a) Before post classification, the selected LLM constructs a model-specific procedure from researcher-authored task materials; the code saves it for reuse. (b) Each routed original post is evaluated anew using that procedure and fixed subject, time-frame, evidence, and alternative-interpretation checks. The dashed arrow denotes procedure reuse, not answer reuse. Discovery is skipped when a retained plan exists. Appendix C provides the detailed workflow and exact prompts; label parsing and Phase 1 fallback follow Algorithm 4.

Algorithm 3 Audited end-to-end evaluation of routed predictions

```

1: Input: reference labels  $y_i$ , Phase 1 labels  $\hat{y}_i^{(1)}$ , route indicators  $r_i$ 
2: Input: parsed Phase 2 labels  $\hat{y}_i^{(2)}$  for routed inputs
3: for each example  $i$  do
4:    $\hat{y}_i^{(E)} \leftarrow \hat{y}_i^{(2)}$  if  $r_i = 1$  and parsing succeeds; otherwise  $\hat{y}_i^{(E)} \leftarrow \hat{y}_i^{(1)}$ 
5: end for
6:  $C \leftarrow \sum_i \mathbf{1}[r_i = 1, \hat{y}_i^{(1)} \neq y_i, \hat{y}_i^{(E)} = y_i]$ 
7:  $I \leftarrow \sum_i \mathbf{1}[r_i = 1, \hat{y}_i^{(1)} = y_i, \hat{y}_i^{(E)} \neq y_i]$ 
8: Report corrected errors  $C$ , introduced errors  $I$ , net corrections  $C - I$ 
9: Report end-to-end metrics over all examples using  $\hat{y}^{(E)}$ 

```

throughout. These are complete prompting protocols: their comparison does not isolate rationale length alone, because label disclosure and call structure also differ.

I. TASK-LEVEL SELF-DISCOVER

SELF-DISCOVER separates *constructing a procedure* from *applying it*, as summarized in Figure 2. Each model generates a plan for the three-class task, stores it, and reuses it across routed posts. This *cached plan* is a saved procedure, not stored post-level answers. Every post still requires a new model response.

Task-specific design. We retain the SELECT-ADAPT-IMPLEMENT framework of SELF-DISCOVER [13] and specialize its inputs and execution guidance to proxy emotion classification. Researchers supply the class policy, 18 emotion-oriented reasoning modules, and fixed evidence-checking instructions; the LLM generates the procedure from those materials. The adaptation addresses distinctions that isolated affect words cannot resolve: whose emotion is expressed, whether distress is current or retrospectively resolved, and whether positive wording describes an actual state rather than

a wish. The contribution is this task-specific specification and its integration with selective re-evaluation, not the invention of task-level planning or caching.

Who constructs the plan, and when. Before classifying a routed post, the code checks for a retained plan for the selected LLM. If none exists, that LLM receives only the researcher-authored task description, policy, and module bank. It does not receive the first post, any other evaluation post, a reference label, or a Phase 1 prediction during discovery. SELECT chooses relevant modules; ADAPT specializes them to the task; IMPLEMENT generates the reusable procedure, which the code stores. If a plan already exists, all three discovery calls are skipped. Llama 2 and Llama 3 generate separate plans; each model reuses its own plan across Reddit and Mixed Emotion.

IMPLEMENT requests at most three one-line JSON steps, each returning to the original text and deferring label selection until the end. This format is requested, not enforced; the realized plans are reproduced in Appendix C. Access to the post is an instruction for subsequent execution, not an input to plan construction.

Per-post execution. The prompt combines the plan, policy, original post, and fixed checks. It asks whose emotion is expressed, whether it is current or retrospectively resolved, and which quotations support or oppose each label. A counterfactual check asks what evidence would favor the second-best label and whether it is present. The model emits a final classification line. SD never receives the Phase 1 label; the evaluator retains it as a fallback if parsing fails.

This division keeps the procedure shared while requiring fresh, post-specific evidence and a new decision for every routed input. The evidence ledger and counterfactual check are researcher-authored execution instructions, not automa-

Algorithm 4 Final task-level SELF-DISCOVER re-evaluation

```

1: Input: model  $g$ ; researcher-authored task, policy, modules, execution
   guidance; routed posts
2: if a retained plan for  $g$  exists then
3:   Load that plan; skip discovery
4: else
5:   Give  $g$  task materials only; no post text or labels
6:    $g$ : SELECT relevant modules and ADAPT them to the task
7:    $g$ : IMPLEMENT the procedure; store it as the plan
8: end if
9: for each routed post  $x$  do
10:  Execute the plan on the original post, without the Phase 1 label
11:  Request source quotes, three-label evidence, and a counterfactual
   check
12:  Parse the final label; retain Phase 1 if parsing fails
13:  Store the response, parsed status, and final prediction
14: end for

```

ically discovered properties of the cached plan. Generated explanations do not establish faithful internal reasoning. Appendix C provides the workflow, templates, and actual plans; Appendix D compares the earlier per-post and final task-level designs without attributing their differences to any single component.

J. INPUT AND OUTPUT POLICY

Reddit Phase 1 uses cleaned title–body text; Phase 2 uses the linked, minimally sanitized original title and selftext. The original-text policy preserves punctuation, negation, and discourse structure but changes the input representation as well as the prediction mechanism. Improvements are therefore attributed to the complete re-evaluation configuration, not solely to reasoning. Long posts follow the notebook’s 6,000-character input policy. Mixed Emotion uses the complete synthetic text rather than a separately lemmatized version.

The three permitted outputs are Depression, Neutral, and Happy. The parser prioritizes the terminal label marker and records unparseable responses. Such responses retain the Phase 1 prediction in the end-to-end evaluation; parsing failure is not automatically counted as a classification error. The remaining final SELF-DISCOVER failures are not regenerated or assigned labels from the reference answer. Label normalization and any recovery of an explicit answer must be applied without consulting the reference label.

VI. EXPERIMENTAL SETUP**A. CLASSIFIER SELECTION AND CALIBRATION**

All classifier configurations use the common 84,000/12,000/12,000/12,000 train/validation/calibration/test partitions, a maximum length of 256 tokens, and a four-trial Bayesian search maximizing validation macro F1. The selected DistilBERT settings are learning rate 4.0368×10^{-5} , batch size 64, an epoch budget of 3, and weight decay 10^{-3} . The epoch budget is a selected training hyperparameter, not a statement that every seed’s best validation checkpoint occurred at that epoch. Seeds 42, 43, and 44 characterize classifier variation; the final downstream run uses seed 42. Appendix Table A1c records the selected settings for all three classifiers.

The downstream export loads the saved reference seed-42 model from this selected configuration; it does not perform a new training run inside the export notebook. Its environment record identifies fp32 evaluation, a 256-token maximum input length, and the same selected hyperparameters. The final model is calibrated with $T^* = 1.480195$ and routed at $\tau^* = 0.70$. The calibration partition has 11,807 accepted and 193 routed posts, accepted-set empirical risk 1.9395%, and one-sided upper bound 2.1615%, below the 5% target. These calibration counts are distinct from the 218 routed Reddit test posts. No Mixed Emotion labels are used for temperature fitting or threshold selection.

B. EXECUTION AND RESOURCE SCOPE

The retained Phase 1 benchmark compares the selected classifier configurations on NVIDIA A100-SXM4-40GB hardware, with fp16 inference, a 256-token cap, and batch size 32. Three timing repetitions are distinct from the three training seeds. These classifier-stage measurements are not measurements of the complete final reasoning pipeline; Appendix Table A1d retains their original measurement scope.

Phase 2 uses chat-tuned Llama 2 7B and instruction-tuned Llama 3 8B with quantized loading and greedy decoding. Direct and final CoT request up to 1,024 new tokens per call. Final SELF-DISCOVER requests up to 1,024 for Llama 2 and 1,536 for Llama 3 in its per-post execution call. Generation budgets therefore remain part of each complete protocol, and Llama 2’s shorter context also constrains multi-turn inference. The appendices disclose template text and nominal execution budgets; a matched end-to-end time and cost comparison is reserved for separate benchmarking rather than inferred from accuracy.

C. EVALUATION AND STATISTICAL SCOPE

Accuracy and macro F1 use all examples, after recombining accepted and routed predictions. Corrected errors C are Phase 1 mistakes made correct by re-evaluation; introduced errors I are previously correct predictions made wrong. The full-set accuracy change in percentage points is $100(C - I)/N$. Wrong-to-wrong changes are not corrections. Routed accuracy, error capture, and accepted-set risk measure different aspects of the pipeline.

All 15 final conditions are aligned by example ID and checked against the stored Phase 1 predictions and reference labels. The checks cover missing or duplicate IDs, parsed outputs, fallback behavior, full-set accuracy, macro F1, class-wise metrics, and corrected/introduced counts. The additional evaluation also validates 777 saved generation calls against their run settings and request hashes. No failed response is regenerated or assigned a reference-derived label.

Two-sided exact McNemar tests compare paired correctness. Against Phase 1, discordant counts are C and I ; between two re-evaluators, they are the numbers correct only under each method. We estimate 95% percentile-bootstrap intervals for accuracy differences with 50,000 multinomial resamples of the paired correctness-difference categories $+1, -1, 0$. This

yields the same distribution as resampling paired rows for that scalar difference. Holm correction uses the 12 main configuration-versus-Phase 1 tests as one family and the three additional-evaluation tests as another. Within each dataset-model combination, the two SD-versus-Direct/CoT tests form a separate family. Appendix E reports unadjusted and adjusted values and all comparison scopes.

These intervals characterize example-level accuracy differences conditional on the fitted models and protocols. They are not confidence intervals for macro F1 or training-seed variability, and they do not account for within-author dependence. The main prompt-development comparisons are exploratory; the additional evaluation applies the frozen final protocols without adapting them during that run.

VII. EXPERIMENTAL RESULTS

A. PHASE 1 PERFORMANCE ON REDDIT DATA

The matched Phase 1 comparison used the same class-balanced 120,000-post sample, prespecified 70/10/10/10 partitions, 12,000-example held-out test set, maximum input length of 256 tokens, four-trial Bayesian search budget, and random seeds 42, 43, and 44 for all three architectures. Table 1 reports predictive performance and inference efficiency for these configurations; training configurations are documented in Appendix Table A1c.

Figure 3 visualizes the same observed three-seed results and their 95% confidence intervals. DistilBERT achieved $96.88 \pm 0.04\%$ accuracy, compared with $95.36 \pm 0.08\%$ for Mistral and $95.17 \pm 0.07\%$ for Llama 2. Macro F1 showed the same pattern. Under this fixed comparison protocol, DistilBERT exceeded Mistral and Llama 2 by 1.52 and 1.71 percentage points in mean accuracy, respectively. The two probes differed by 0.19 points; their relative ordering is not the basis of the operational choice.

DistilBERT had the highest mean score and approximately 67 million parameters, compared with about seven billion for each comparator. Under the tested configurations and shared budget, its predictive performance and measured inference efficiency support its selection as the first-stage classifier. Depression-Happy confusions accounted for 72.8% of DistilBERT errors, 75.6% of Llama 2 errors, and 74.3% of Mistral errors across pooled seed predictions; Neutral F1 exceeded 98% for all three. Pooling summarizes error composition, not 36,000 independent test examples. Small probe seed deviations also reflect the frozen backbone and should not be interpreted as equivalent training variability across adaptation methods.

The final linked DistilBERT run achieves 96.9167% accuracy and 96.9166% macro F1 on Reddit. This run uses the same selected hyperparameter configuration as the three-seed comparison. Its calibration and routing outputs, rather than those of a separately configured model, define every downstream condition.

B. CLASS-LEVEL ERROR STRUCTURE

Figure 4 shows the final linked classifier's errors on both datasets. Cell counts and row-normalized percentages distin-

guish the balanced Reddit test from the smaller stress test. These are the final Phase 1 predictions, not pooled three-seed counts or re-evaluation outputs.

C. CALIBRATION AND ROUTING

Temperature scaling reduces all four calibration diagnostics (Table 2). The fixed test router selects 218 of 12,000 Reddit posts and 86 of 300 Mixed Emotion examples. The difference in routing volume is an observed response to the stress-test inputs, not a dataset-specific threshold adjustment.

On Reddit, the routed subset contains 98 of the 370 Phase 1 errors (26.49% error capture). Its error rate is 44.95%, compared with 3.08% over the full set, an approximately 14.58-fold enrichment. The accepted subset still contains 272 errors, beyond the scope of routed-only correction. On Mixed Emotion, routing captures 28 of 49 errors (57.14%); 21 errors remain accepted. Routed-set Phase 1 accuracy is 55.05% on Reddit and 67.44% on Mixed Emotion.

D. END-TO-END MODEL AND PROMPTING RESULTS

Table 4 reports the complete final comparison. Llama 3 SELF-DISCOVER produces the largest observed Phase 1 gain on both datasets: 34 net corrections on Reddit and 27 on Mixed Emotion, reaching 97.2000% and 92.6667% accuracy. It also leads the six configurations in accuracy and macro F1 on each dataset. On Mixed Emotion, SD exceeds both alternatives within each model after Holm adjustment; for Llama 3, adjusted $p = 0.0020$ versus Direct and $p < 0.0001$ versus CoT. On Reddit, its six-prediction lead over Llama 3 CoT is not statistically significant, nor is its difference from Direct (Holm $p = 0.3616$ and 0.2163).

E. CORRECTION AND HARM

On Reddit, Llama 3 SELF-DISCOVER corrects 72 errors and introduces 38, producing 34 net corrections. Direct and CoT yield 23 and 28 net corrections, respectively. Llama 2 yields smaller gains: Direct has nine net corrections, CoT one, and SELF-DISCOVER three. For Llama 2, Direct, CoT, and SELF-DISCOVER yield nine, one, and three net corrections, showing a model-protocol interaction.

On Mixed Emotion, Llama 3 SELF-DISCOVER corrects 28 errors and introduces one. Total errors decrease from 49 to 22, a 55.10% relative reduction, and accuracy increases by 9.00 percentage points. The single introduced error indicates strong preservation of initially correct labels within this controlled 300-example stress test. Llama 2 SELF-DISCOVER also improves the baseline (17 net corrections), whereas its Direct and CoT configurations introduce more errors than they correct. The pattern supports evaluating both correction yield and damage to initially correct predictions.

F. ADDITIONAL EVALUATION OF THE FIXED PIPELINE

On the separate 9,000-post set, Phase 1 correctly classifies 8,782 posts (97.5778%) and routes 111 (1.23%). The routed subset contains 48 of the 218 Phase 1 errors: its error rate is 43.24%, compared with 2.42% overall. Thus, the fixed

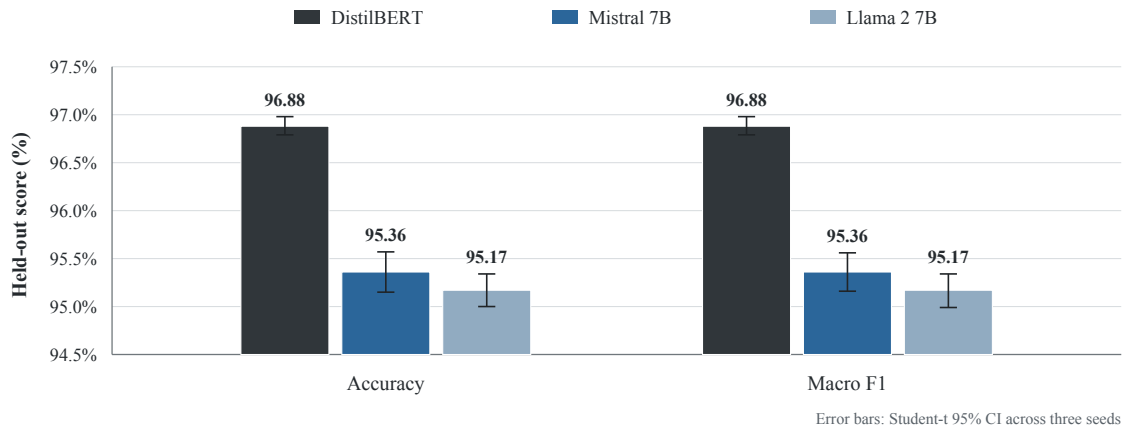
TABLE 1. Matched Phase 1 predictive performance and inference efficiency

(a) Predictive performance on the common 12,000-post test set				
Model	Accuracy, mean $\pm$ SD	Accuracy 95% CI	Macro F1, mean $\pm$ SD	Macro F1 95% CI
DistilBERT	96.88 $\pm$ 0.04%	[96.79, 96.98]%	96.88 $\pm$ 0.04%	[96.79, 96.98]%
Mistral 7B	95.36 $\pm$ 0.08%	[95.15, 95.57]%	95.36 $\pm$ 0.08%	[95.16, 95.56]%
Llama 2 7B	95.17 $\pm$ 0.07%	[95.00, 95.34]%	95.17 $\pm$ 0.07%	[94.99, 95.34]%

Note. Panel (a) reports percentages over training seeds 42, 43, and 44. Confidence intervals are two-sided Student-*t* intervals across the three seed-level results. The comparison evaluates a bounded operational screening protocol; it does not estimate fully tuned 7B performance.

(b) Inference on NVIDIA A100-SXM4-40GB			
Model	Throughput (posts/s)	Single-post p50 (ms)	Peak memory (GB)
DistilBERT	4,959.5 $\pm$ 15.5	4.80 $\pm$ 0.03	0.29
Mistral 7B	51.7 $\pm$ 0.0	36.62 $\pm$ 0.06	15.22
Llama 2 7B	55.5 $\pm$ 0.0	34.05 $\pm$ 0.10	14.05

Note. Panel (b) reports mean $\pm$ SD across three timing repetitions, distinct from the training seeds in panel (a). Throughput uses batch size 32 and all 12,000 test posts; p50 is the median latency for 512 posts processed individually. All models use fp16 and a 256-token cap. Memory refers to batched inference. An SD displayed as 0.0 reflects rounding. Appendix Table A1d provides p95 latency and the full protocol.

**FIGURE 3. Observed Phase 1 classifier performance on the common 12,000-example held-out Reddit test set. Columns show three-seed means and whiskers show Student-*t* 95% confidence intervals for accuracy and macro F1. These intervals describe training-run variation, not repeated timing measurements. The vertical axis is truncated to make the between-model differences legible.****TABLE 2. Final DistilBERT calibration diagnostics (calibration split)**

Score	Temperature	NLL	Brier	ECE	Adaptive ECE
Raw MSP	1.0000	0.0933	0.0443	0.0176	0.0171
Temperature-scaled MSP	1.4802	0.0795	0.0416	0.0064	0.0128

threshold again selects an error-enriched subset while retaining 98.77% coverage.

Table 5 reports the fixed-protocol results. SELF-DISCOVER yields 17 net corrections, increases accuracy from 97.5778% to 97.7667%, and significantly improves over Phase 1 by 0.1889 percentage points (95% CI [0.0444, 0.3333]; Holm $p = 0.0412$). Direct and CoT each yield 13 net corrections. SELF-DISCOVER retains the highest observed accuracy, although its four-correct-prediction lead over either alternative is not statistically significant (Holm $p = 0.9614$ for both comparisons). There are two Direct parsing failures

and none for CoT or SD; all use the same fallback rule. Appendix G provides the fixed settings and class-level results.

G. CONDITIONAL CORRECTION OPPORTUNITY

With the router held fixed, a conditional oracle corrects every routed error without changing any initially correct prediction. Its full-set ceiling is $\text{Acc}_{P1} + E_R/N$, where E_R is the number of routed Phase 1 errors. The ceiling is 97.7333% on Reddit and 93.0000% on Mixed Emotion. Final Llama 3 SELF-DISCOVER realizes $34/98 = 34.69\%$ and $27/28 = 96.43\%$ of these respective net-correction opportunities. On the additional 9,000 posts, the corresponding ceiling is 98.1111%

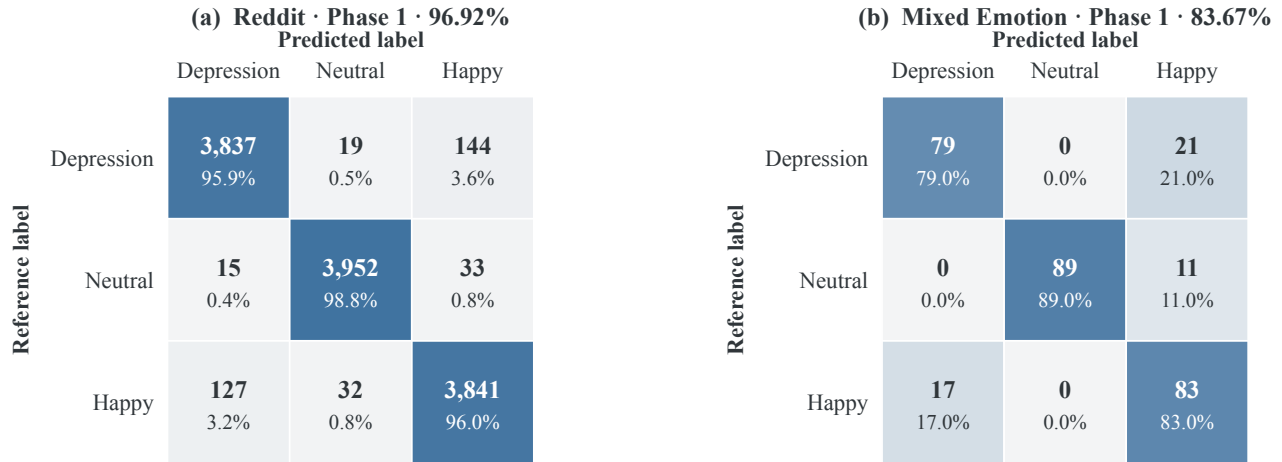


FIGURE 4. Final Phase 1 confusion matrices on Reddit ($N = 12,000$) and Mixed Emotion ($N = 300$). Rows are reference labels; columns are predictions. Each cell shows a count and its percentage within the reference class. Both panels use the same row-normalized color scale.

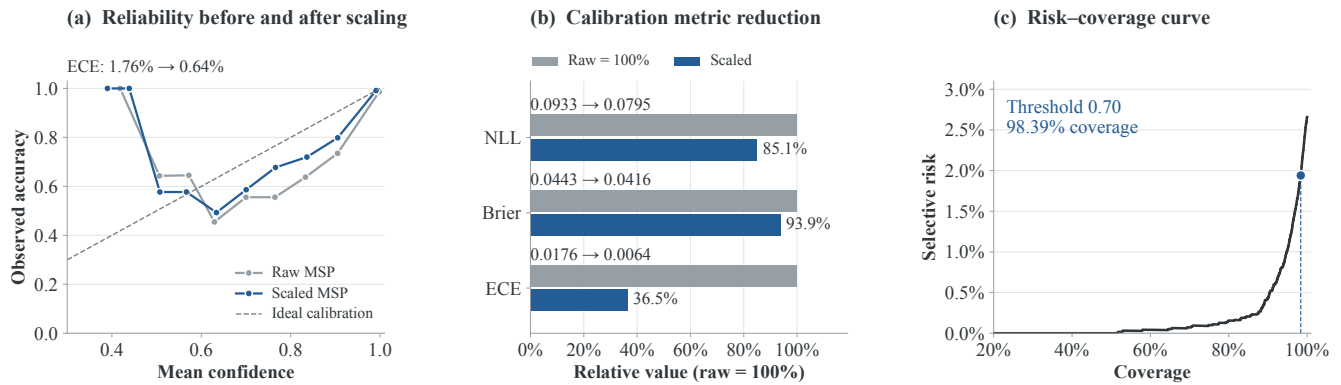


FIGURE 5. Calibration diagnostics for the final linked DistilBERT model. Reliability, relative metric reduction, and risk-coverage curves use the calibration split; the selected point is not selected on test outcomes.

TABLE 3. Fixed-policy routing outcomes; rates are percentages

Dataset	N	Routed	Route rate	Coverage	Error capture	Routed error	Accepted risk
Reddit	12000	218	1.82	98.18	26.49	44.95	2.31
Mixed Emotion	300	86	28.67	71.33	57.14	32.56	9.81

and SD realizes $17/48 = 35.42\%$ of the net-correction opportunity. These are descriptive ceilings under a fixed router, not achievable performance guarantees or alternative tuned baselines.

VIII. DISCUSSION

A. AN EFFICIENT FIRST STAGE AND A SELECTIVE SECOND STAGE

The linked experiment separates three decisions: selecting a practical classifier, routing uncertain posts, and choosing a re-evaluation protocol. DistilBERT retains strong three-seed predictive performance and a measured classifier-stage inference advantage under the bounded Phase 1 comparison. The final run then supplies one consistent set of predictions and confidence scores for every Phase 2 condition. Classifier

means and downstream single-run scores answer different questions without requiring different hyperparameter configurations.

B. SELECTION AND CORRECTION ARE DIFFERENT OUTCOMES

Confidence routing creates an error-enriched correction opportunity; the re-evaluator determines the resulting net gain. Llama 2 CoT is nearly neutral on Reddit and harmful on Mixed Emotion, whereas final SELF-DISCOVER with Llama 3 improves both sets. On the stress test, searching for evidence for and against all three classes and distinguishing present distress from resolved or attributed events are plausible contributors to the observed pattern. The experiment evaluates the complete

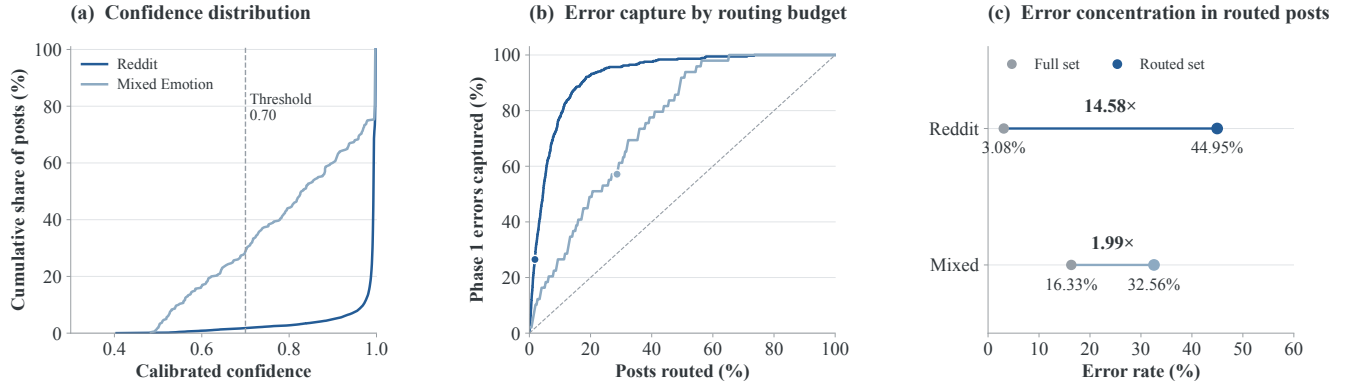


FIGURE 6. Fixed-policy routing concentration. (a) Cumulative calibrated-confidence distributions. (b) Errors captured when posts are ordered from lowest to highest confidence; dots identify the fixed threshold and the diagonal is a proportional-capture reference. (c) Full-set and routed error rates, with enrichment ratios. Curves describe the evaluation sets; they are not used to select the threshold.

TABLE 4. Final full-set outcomes on 12,000 Reddit posts and 300 Mixed Emotion examples. Accuracy and macro F1 are percentages; change is relative to Phase 1 in percentage points. C counts corrected errors and I newly introduced errors; $C - I$ is their net difference. Unparsed responses retain Phase 1 predictions.

Dataset	Model	Protocol	Accuracy	Macro F1	C	I	$C - I$	Change
Reddit	DistilBERT	Phase 1	96.9167	96.9166	—	—	—	—
Reddit	Llama 2	Direct	96.9917	96.9898	45	36	+9	+0.0750
Reddit	Llama 2	CoT	96.9250	96.9240	48	47	+1	+0.0083
Reddit	Llama 2	SELF-DISCOVER	96.9417	96.9391	52	49	+3	+0.0250
Reddit	Llama 3	Direct	97.1083	97.1056	68	45	+23	+0.1917
Reddit	Llama 3	CoT	97.1500	97.1471	63	35	+28	+0.2333
Reddit	Llama 3	SELF-DISCOVER	97.2000	97.1990	72	38	+34	+0.2833
Mixed Emotion	DistilBERT	Phase 1	83.6667	84.0005	—	—	—	—
Mixed Emotion	Llama 2	Direct	76.3333	75.1627	11	33	-22	-7.3333
Mixed Emotion	Llama 2	CoT	79.0000	78.9298	19	33	-14	-4.6667
Mixed Emotion	Llama 2	SELF-DISCOVER	89.3333	89.3452	22	5	+17	+5.6667
Mixed Emotion	Llama 3	Direct	89.3333	89.3803	26	9	+17	+5.6667
Mixed Emotion	Llama 3	CoT	84.6667	84.6760	15	12	+3	+1.0000
Mixed Emotion	Llama 3	SELF-DISCOVER	92.6667	92.6966	28	1	+27	+9.0000

TABLE 5. Additional 9,000-post evaluation with the final Llama 3 protocols fixed. Accuracy and macro F1 are percentages. Change and its 95% interval are in percentage points relative to Phase 1; Holm p adjusts the three comparisons with Phase 1.

Protocol	Accuracy	Macro F1	C	I	Change	Paired 95% CI	Holm p
Phase 1	97.5778	97.5821	—	—	—	—	—
Direct	97.7222	97.7242	32	19	+0.1444	[-0.0111, 0.3000]	0.1320
CoT	97.7222	97.7234	28	15	+0.1444	[0.0000, 0.2889]	0.1320
SELF-DISCOVER	97.7667	97.7688	30	13	+0.1889	[0.0444, 0.3333]	0.0412

prompt package; it does not establish the separate causal effect of each instruction or demonstrate that the generated rationale faithfully reveals the model's computation.

C. WHAT THE MODEL-PROTOCOL COMPARISON SHOWS

Llama 3 SELF-DISCOVER has the highest observed accuracy and macro F1 on all three evaluation sets and significantly improves final accuracy over Phase 1 under the stated Holm adjustments. Within-model protocol comparisons answer a narrower question: SD's advantage over Direct and CoT is

supported on Mixed Emotion, while the smaller differences on the Reddit evaluations remain uncertain. Moreover, the 0.2833-point Reddit gain uses a 1.82% routing rate, whereas the 9.00-point Mixed Emotion gain uses 28.67%. These are different correction opportunities, not directly comparable effect sizes. Likewise, one introduced stress-test error versus 38 Reddit errors warrants dataset-specific interpretation rather than a universal reliability claim.

The methodological value of the adaptation is to specify what evidence a selected post should be checked against, rather

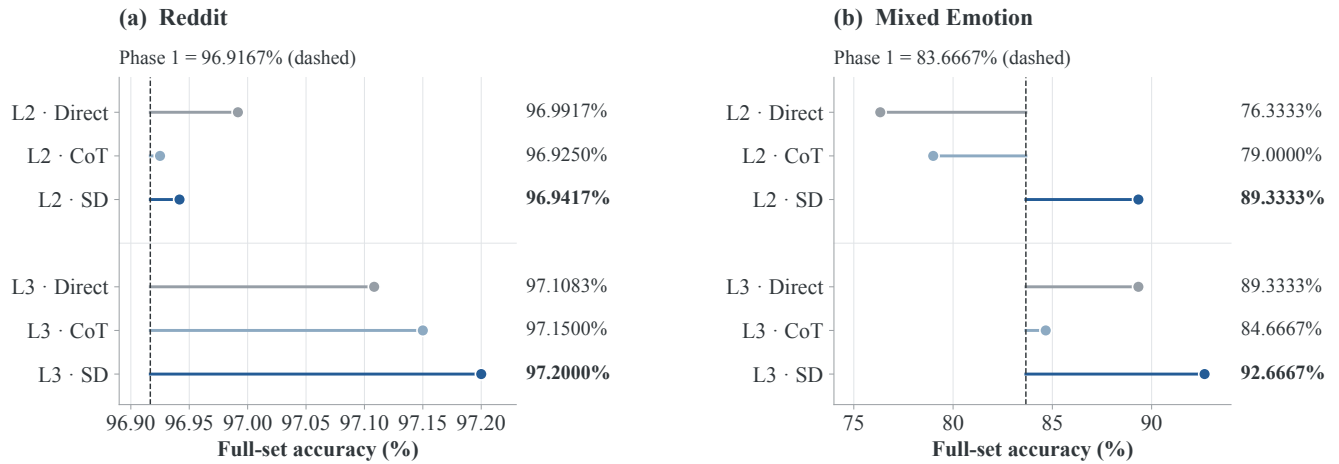


FIGURE 7. Full-set accuracy for the final six configurations. Points show observed accuracy; horizontal segments connect each point to the linked Phase 1 baseline (dashed line). The two panels use different axis ranges. SD denotes the final task-level compact-plan SELF-DISCOVER protocol.

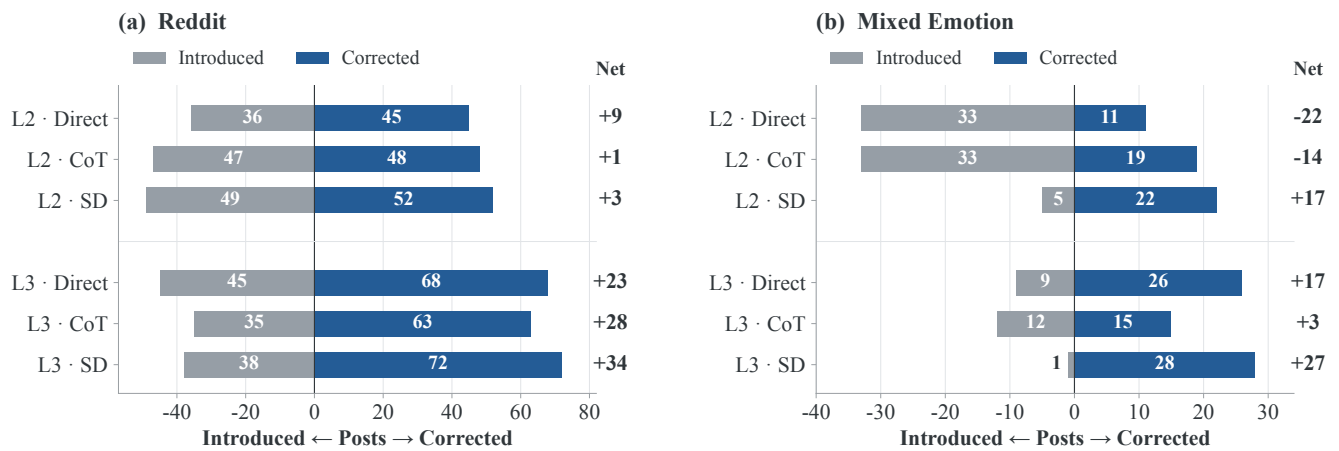


FIGURE 8. Corrected (blue, right) and introduced (gray, left) errors for each final configuration. Introduced counts are plotted to the left for comparison, not as negative error counts. Right-hand labels show net corrections. The panels use different count scales.

than merely requesting a longer explanation. Emotional subject and time-frame checks address attribution and trajectory, while the three-label ledger and counterfactual check make competing interpretations explicit. The observed gains support this complete configuration within the evaluated settings; they do not identify which instruction caused the gain. Appendix D shows that the final design improves on the earlier independent-countercheck variant for Llama 3 on both sets, but not for every model–dataset combination.

The final model-by-protocol comparison is more informative than contrasting different models with different prompts only. Nevertheless, different context windows, generation budgets, and exposure to the Phase 1 label remain part of the evaluated configurations. Llama 3’s advantage is therefore a configuration-level finding, not an isolated causal estimate of model generation or parameter count. Likewise, Direct is a label-only re-evaluation baseline: its benefit does not prove that an explicit reasoning trace is necessary.

IX. LIMITATIONS AND FUTURE WORK

Subreddit-derived, sentiment-filtered labels are proxy constructs and can reward topical or community-specific shortcuts. Random post-level splits do not establish author-disjoint or temporal independence. The Mixed Emotion data are synthetic, scenario-balanced, and not independently expert-labeled; their results should not be generalized to naturally occurring emotional ambiguity or clinical depression.

The main Reddit and Mixed Emotion cases informed prompt development, so their configuration rankings are exploratory and the reported corrections do not adjust for prompt selection. The separate 9,000-post evaluation applies the final Llama 3 protocols and retained plan without changes during evaluation, providing an additional same-source check. It does not establish author-disjoint, temporal, cross-platform, or clinical generalization, and its small between-protocol differences remain uncertain.

Reddit re-evaluation uses richer original text than the cleaned Phase 1 input. This is a deliberate system design

(a) Reddit - final SD				(b) Mixed Emotion - final SD					
		Predicted label					Predicted label		
		Depression	Neutral	Happy			Depression	Neutral	Happy
Reference label	Depression	3,851 96.3%	24 0.6%	125 3.1%	Reference label	Depression	82 82.0%	0 0.0%	18 18.0%
	Neutral	13 0.3%	3,954 98.9%	33 0.8%		Neutral	0 0.0%	97 97.0%	3 3.0%
	Happy	104 2.6%	37 0.9%	3,859 96.5%		Happy	0 0.0%	1 1.0%	99 99.0%

FIGURE 9. Final Llama 3 SELF-DISCOVER confusion matrices, reconstructed from all end-to-end predictions. Rows are reference labels and columns are predictions; counts and within-class percentages share the Phase 1 figure's scale. Unparsed routed responses retain the initial label.

but prevents attributing the entire gain to prompting alone. Residual parsing failures retain Phase 1 predictions; their presence and the rule must be considered when comparing protocols. Final SELF-DISCOVER has 21 parsing failures for Llama 2 on Reddit and none on Mixed Emotion, and two and one, respectively, for Llama 3. The additional Llama 3 SD evaluation has no parsing failures. The 21 Llama 2 Reddit fallbacks are part of its measured configuration-level performance rather than excluded cases. No unresolved failure is manually relabeled from its reference answer.

The current experiments do not quantify repeated-generation variation for all final configurations or supply a fully matched end-to-end cost benchmark. Greedy decoding does not guarantee bitwise equality across hardware and software environments. Independent expert review, broader evaluation, and generation-cost accounting are useful extensions, rather than reasons to repeat the completed Phase 1 model comparison.

X. CONCLUSION

A unified DistilBERT-to-LLM pipeline connects efficient classifier selection, calibrated routing, and selective original-text re-evaluation. Across the main Reddit test, the Mixed Emotion stress test, and the additional 9,000-post evaluation, Llama 3 SELF-DISCOVER converts the routed correction opportunity into 34, 27, and 17 net corrections, respectively, and significantly improves accuracy over Phase 1 under the stated Holm adjustments. It achieves the highest observed accuracy and macro F1 among the evaluated configurations on all three sets. Paired comparisons further show that SD exceeds Direct and CoT on Mixed Emotion within both models, while the smaller between-protocol differences on Reddit remain unresolved. The results support the task-specific reusable plan and evidence checks as an effective complete re-evaluation configuration and demonstrate that model and protocol choice determine how routing translates into final gains. These findings establish value for proxy-label text

classification; clinical use requires separate validation and governance.

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APPENDIX A ADDITIONAL METHODOLOGICAL DETAILS

Appendix Table A1 summarizes the main variables retained from the Reddit dataset during data construction and preprocessing. Appendix Table A1b complements the schema description with class-balanced examples from the controlled Mixed Emotion stress test.

Appendix Table A1. Description of Reddit dataset variables

Variable	Description
Subreddit	The subreddit name where the post was published, indicating the context, topic, and implicit emotional milieu of the post.
Author	The anonymized Reddit username of the post's author.
Over_18	A binary flag indicating whether the post was tagged as adult content.
Title	The title of the Reddit post, often providing a brief summary of the content.
Selftext	The main body of the post, containing the full text written by the user, if available.
Created_UTC	The timestamp, in UTC, indicating when the post was created; this variable can support temporal analysis.
Title_with_selftext	The concatenation of the title and body text, representing the complete textual content of the post before cleaning.
Title_with_selftext_cleaned	The processed text after cleaning, used as the primary input for modeling.
Polarity	A document-level TextBlob polarity score computed by passing the cleaned concatenated title and body to <code>TextBlob(...).sentiment.polarity</code> ; it is not a manually averaged token score.
Class_group	The target label denoting the emotional class: Depression, Neutral, or Happy.

The examples below make the stress-test construct observable: the target is determined by the dominant full-text trajectory rather than by the presence of any single positive or negative word.

Appendix Table A1b. Representative inputs from the Mixed Emotion stress test

Example ID	Target	Actual input text	Dominant-label rationale
MEV2_DEP_001	Depression	During a school day, I noticed a kind message from a friend, and ordinary tasks still needed to be finished. Even with those ordinary or positive moments, there was a heavy sadness returning in the background. I kept rereading the same message without knowing how to answer, especially before breakfast. By the end, the positive parts felt brief, and the heavier sadness was still the feeling I carried with me.	Positive and routine details coexist with the negative cues, but the ending preserves unresolved sadness as the dominant trajectory.
MEV2_NEU_001	Neutral	While reviewing a class registration page, I noticed that one course appeared in two sections with different room numbers. I copied both entries into a note, checked the department page, and left the final choice blank until the schedule page is updated. The post is mainly about checking inconsistent information and does not state a strong personal reaction.	A minor procedural ambiguity is present, but the post remains factual and low in affective intensity.
MEV2_HAP_001	Happy	While dealing with a school day, I could still feel a heavy sadness returning in the background. At the same time, there was a kind message from a friend, and ordinary tasks still needed to be finished. I kept rereading the same message without knowing how to answer, especially before breakfast. Even with the mixed parts, I ended the day feeling relieved, quietly happy, and more connected than before.	The input contains sadness and routine details, but its explicit final resolution is relief, happiness, and renewed connection.

Note. These are unmodified input texts from the final 300-example controlled stress-test dataset. They were not used for Phase 1 training, temperature fitting, or threshold selection. The table illustrates co-occurring cues and final-trajectory labeling; it is not intended to establish population prevalence or clinical validity.

Appendix Table A1c. Sweep-selected Phase 1 configurations

Model	Training	Learning rate	Batch	Epochs	Weight decay
DistilBERT	Full fine-tuning	4.037×10^{-5}	64	3	10^{-3}
Llama 2 7B	Linear probe	4.151×10^{-4}	32	3	10^{-4}
Mistral 7B	Linear probe	2.399×10^{-4}	64	2	10^{-4}

Note. Each row reports the configuration selected by a four-trial Bayesian sweep that maximized validation macro F1. "Epochs" is the training budget selected by the sweep; each complete configuration was subsequently evaluated with seeds 42, 43, and 44. The final downstream DistilBERT run uses the selected configuration in this table, with temperature 1.480195 and threshold 0.70. The single-run downstream score is not the three-seed mean.

Appendix Table A1d. Phase 1 computational measurements

(a) One-off classifier development

Model	Feature extraction (GPU-h)	Search (GPU-h)	Three-seed fitting (GPU-h)	Build total (GPU-h)	Peak memory: extraction / fitting (GB)
DistilBERT	—	0.32	0.27	0.59	— / 4.35
Mistral 7B	0.67	0.02	0.003	0.70	29.11 / 1.11
Llama 2 7B	0.64	0.03	0.01	0.68	27.01 / 1.11

(b) Repeated inference measurements

Model	Batched time (min)	Single-post p95 (ms)	Single-post peak memory (GB)	GPU-h per 1,000 posts	Batched GPU energy (Wh)
DistilBERT	0.040	5.11 ± 0.12	0.16	0.000056	0.16
Mistral 7B	3.87	38.37 ± 0.18	14.28	0.005376	25.5
Llama 2 7B	3.61	36.23 ± 0.40	13.27	0.005008	23.8

(c) Complete measured run, including evaluation and repeated benchmarks

Model	Total GPU time (h)	GPU-board energy (Wh)	Batched GPU utilization (%)
DistilBERT	0.61	124.6	57.0
Mistral 7B	0.91	335.7	99.7
Llama 2 7B	0.87	310.8	99.7

Protocol. All three configurations were measured on NVIDIA A100-SXM4-40GB accelerators in the same software environment. Training used bf16 and the inference benchmark used fp16. The input-length cap was 256 tokens. Batched inference used batch size 32 on all 12,000 held-out posts; single-post latency used 512 posts at batch size one. Each inference benchmark was repeated three times after three warm-up batches. Table 1(b) reports mean throughput, mean p50 latency, and their across-repetition SDs; panel (b) above reports mean p95 latency and its SD. Displayed zero SDs result from rounding, not an assumption of zero variation.

Timing and resource accounting. CUDA was synchronized before and after each timed region, including host-to-device copies; this is a model-inference benchmark rather than full application latency. Non-padding input tokens were counted. NVML sampled utilization and board power at 1 Hz, and energy was computed by trapezoidal integration of board-power samples. Energy covers the GPU board only; the short DistilBERT timing windows limit the temporal resolution of its energy and utilization estimates. Batched time and energy in panel (b) describe one repetition, not the sum of three. The GPU-hours per 1,000 posts are normalized from batched throughput. Values are rounded independently.

Build and evaluation scope. Probe feature extraction processed 108,000 posts before fitting cached-feature linear heads. The three-seed fitting and fitting-memory entries therefore exclude the backbone, whose cost and peak memory appear separately in panel (a). The search used four trials per model. Cached-feature head scoring is not classifier inference throughput; neither those timings nor the DistilBERT training-pipeline scoring rate are used in Table 1(b). Panel (c) includes search, fitting, scoring, and all timing repetitions, and is not a per-deployment workload estimate. Monetary conversion and Phase 2 generation costs are not included.

APPENDIX B FINAL PROMPTING PROTOCOLS

The following templates are extracted from the executed notebook code. Placeholder fields are substituted at runtime. The shared policy is identical in the supplied final SD and Llama 3 baseline templates; the final Llama 2 CoT instruction sequence matches the Llama 3 sequence. Direct discloses the Phase 1 label immediately, CoT discloses it after an independent assessment, and final SD never discloses it. No reference label is inserted into any prompt.

Appendix Table B1. Final re-evaluation protocols

Protocol	Information supplied	Processing structure	Scored output
Direct	Original post, shared policy, Phase 1 label	One label-only generation call	One parsed label; otherwise Phase 1 fallback
CoT	Original post; Phase 1 label revealed after independent assessment	Five dialogue calls, including acknowledgment and text-response turns	Terminal label from the final response; otherwise fallback
SELF-DISCOVER	Original post, policy, cached model-specific plan; no Phase 1 label	Task-level discovery, followed by one execution call per routed post	Terminal label after evidence and counterfactual checks; otherwise fallback

The sections below reproduce the literal prompt text. Placeholder names remain as written in the executable templates.

A. SHARED CLASSIFICATION POLICY

Classification Policy:

- Depression: unresolved sadness, hopelessness, emotional distress, emotional exhaustion, withdrawal, self-devaluation, or a clearly negative overall trajectory is dominant.
- Neutral: the text is mainly factual, routine, balanced, informational, or emotionally mild, without a dominant positive or distress-related state.
- Happy: happiness, relief, gratitude, accomplishment, fulfillment, or a clearly positive resolution is dominant.

Assess the dominant emotional meaning of the full text. For mixed or shifting emotions, consider the overall trajectory and final takeaway. Do not decide from isolated words, and do not make clinical diagnoses or treatment recommendations.

B. DIRECT RE-EVALUATION

One user message is formed by inserting the policy, original text, and Phase 1 label into the following template. No explicit rationale is requested.

You are an expert annotator for research-oriented, non-clinical emotion classification.

{policy}

Text:

{text}

The Phase 1 classifier predicted: {phase1_label}

Select the final label that best represents the text. Return no rationale and end with exactly one line in this form:

Final label: Depression

or Final label: Neutral

or Final label: Happy

C. CHAIN-OF-THOUGHT DIALOGUE

The following system message and user instructions are sent in order. Assistant responses are retained in the conversation. After the first user instruction the model generates an acknowledgment; after the text is provided it generates a text response; then independent analysis, comparison, and final-decision calls follow. There are five generation calls per routed post, not one.

System message

You are an expert annotator for research-oriented, non-clinical emotion classification. Assist in analyzing emotions in text data. Do not make a clinical diagnosis, infer a medical condition, or provide treatment advice.

Initial user instruction

I will first provide a piece of text. Independently assess its emotional content before comparing it with a Phase 1 AI-generated label. The only permitted labels are Depression, Neutral, and Happy.

Post input

Text:

{phase2_original_text}

Independent analysis

Independently analyze the text. State the dominant emotion and textual evidence before considering the Phase 1 label.

{policy}

Use only Depression, Neutral, or Happy for your provisional label.

Comparison with Phase 1

The Phase 1 classifier predicted: {phase1_label}

Compare that prediction with your independent assessment. Confirm it only when it is supported by the dominant emotional meaning of the full text. Otherwise, explain the correction using textual evidence only.

Final decision

Provide the final Phase 2 decision. Use only one exact label: Depression, Neutral, or Happy. Do not use synonyms or additional labels. End the response with exactly one line: Final label: [label]

APPENDIX C FINAL SELF-DISCOVER TEMPLATES

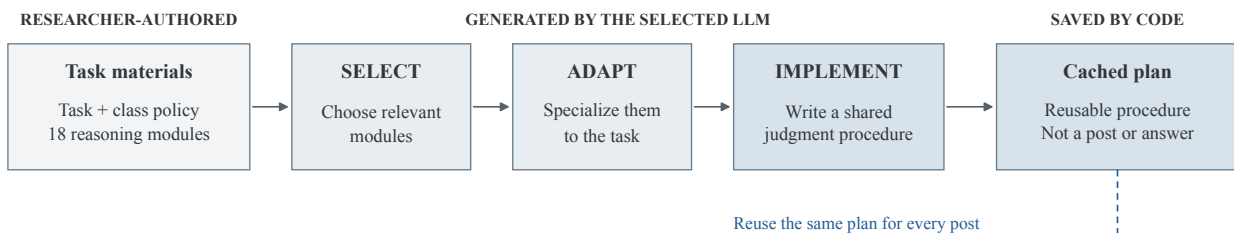
This appendix documents the final task-level SELF-DISCOVER protocol (artifact identifier v6c). Read it in two parts: *discovery* generates and stores a model-specific procedure, whereas *execution* applies that procedure to each routed original post. Figure C1 separates these scopes. The source templates below are reproduced verbatim; the explanatory paragraphs describe how they connect, rather than modifying the prompts.

The chronology and authorship are distinct. Researchers first write the task description, class policy, module bank, and execution guidance. When no retained plan exists, the selected LLM performs SELECT, ADAPT, and IMPLEMENT using the task materials only; the code saves its generated procedure before classifying any post. No individual post or reference label enters these discovery calls. Only then does the LLM receive a routed original post together with the saved procedure and fixed execution guidance. A retained plan bypasses all three discovery calls.

The stored plan is a reusable text artifact, not a set of cached predictions. Each model uses its own plan for both datasets. Reusing it does not mean learning a procedure from the first evaluation post, reusing that post's answer, updating model weights, or bypassing per-post inference.

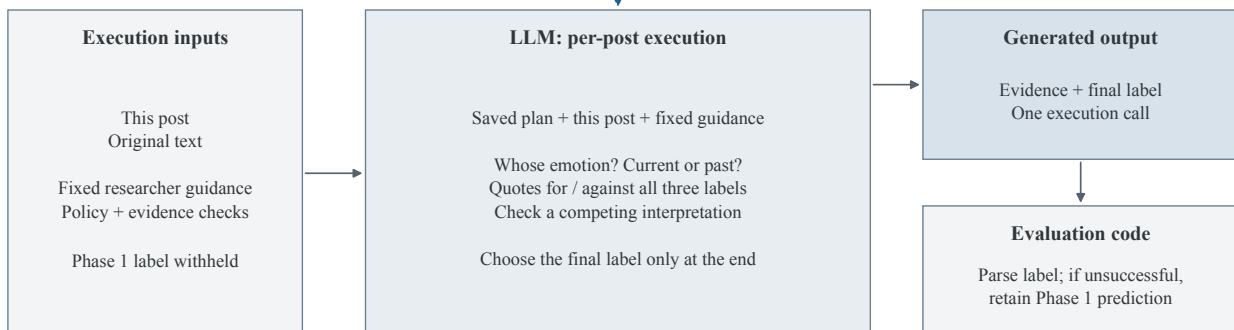
(a) Before post classification: construct the plan if no saved plan exists

No evaluation post or reference label is supplied during SELECT, ADAPT, or IMPLEMENT.



(b) For each routed post: the LLM generates a fresh analysis and label

Post text first enters here. An existing plan bypasses all three discovery calls in (a).



Llama 2 and Llama 3 each generate their own plan; each plan is reused for Reddit and Mixed Emotion.

Appendix Figure C1. Final task-level SELF-DISCOVER workflow, separating authorship, timing, and input scope. (a) Researchers supply task materials; before post classification, the selected LLM generates a plan through SELECT, ADAPT, and IMPLEMENT if no retained plan exists. No evaluation post or reference label enters discovery. (b) The same LLM applies the saved plan and fixed researcher-authored checks to each routed original post, generating a fresh response. The dashed arrow denotes plan reuse, not reuse of an answer. Parsing and fallback are code operations, not additional LLM calls.

Appendix Table C1. Final SELF-DISCOVER stages

Stage	Input	Requested operation	Retained output
SELECT	Task policy and 18 domain modules	Select task-relevant reasoning operations	Selected modules
ADAPT	Task description and selected modules	Specialize operations to the three-class task	Adapted modules
IMPLEMENT	Adapted modules	Request a compact plan of at most three one-line steps	Actual model-generated plan, cached without a conformance validator
Execution	Original post, policy, cached plan	Compare evidence for all three labels and check an alternative interpretation	Response text, parsed label, and failure status

A. TASK DESCRIPTION AND MODULE BANK

Purpose and scope. These are researcher-authored inputs to model-level discovery, not an individual evaluation post. The task description defines the three-label annotation problem, and the module bank supplies 18 candidate operations concerning emotional subject, time frame, trajectory, topic, and competing evidence. The policy placeholder expands to the shared classification policy in Appendix B. The title–body wording is retained verbatim in the Mixed Emotion run, whose supplied input is the synthetic post text.

Task: research-oriented, non-clinical emotion annotation of English social-media posts.

Each instance is one post, consisting of a title and a body, written by a single author. An annotator must assign exactly one label from Depression, Neutral, Happy, according to the following policy.

{policy}

The posts are real, user-written text. They vary greatly in length. A single post may cover several topics, quote or paraphrase other people, ask a question, give advice, report factual information, describe a past episode that has since resolved, or mix positive and negative cues within the same paragraph. The annotator must decide the dominant emotional meaning of the post as a whole, and must be able to point to the text that supports the decision.

No clinical diagnosis, medical inference, or treatment advice is produced. The output is a research annotation label only.

- 1 Separate the writer’s own current emotional state from emotions they attribute to other people, to quoted text, or to hypothetical situations.
- 2 Locate the emotional stance the post opens with and the emotional stance it closes with, and state explicitly whether they differ.
- 3 Determine the final emotional takeaway the writer leaves the reader with, instead of averaging every emotional word in the post.
- 4 Distinguish a past state the writer describes as changed or resolved from a state that is still ongoing at the time of writing.
- 5 Decide whether the post is primarily informational, procedural, advice-giving, or a question about a topic, rather than an expression of the writer’s own current feelings.
- 6 Check for irony, sarcasm, dark humour, or understatement that inverts the literal sentiment of a phrase.
- 7 Weigh explicit emotional statements against implicit signals such as sleep, appetite, isolation, disrupted routine, or loss of interest.
- 8 Identify markers of resolution: relief, gratitude, recovery, accomplishment, reconnection, or a plan the writer feels good about.
- 9 Identify markers of unresolved distress: hopelessness, exhaustion, self-devaluation, numbness, withdrawal, or an appeal for help that the post never resolves.
- 10 Judge emotional intensity, separating strong affect from mild, passing, or conventional affect.
- 11 Check whether the post seeks support or asks a question, and whether the surrounding text is distressed or matter-of-fact about it.
- 12 Separate discussing mental health as a subject, or reporting facts about it, from the writer expressing their own mental state.
- 13 For long or multi-topic posts, identify which segment carries the writer’s main point and weight that segment above digressions.
- 14 Rank all three candidate labels for this post and name which label is second-best, rather than stopping at the first label that seems to fit.
- 15 State the single strongest piece of textual evidence for each of the three labels before choosing between them.
- 16 Test the leading label by asking what would have to appear in the text for a different label to be correct, then check whether it appears.
- 17 Check whether the tone is measured and observational rather than emotionally invested, which can occur even when the subject matter is heavy.
- 18 Check whether apparent positivity is aspirational, hypothetical, conventional politeness, or directed at someone else, rather than reflecting the writer’s own current state.

B. SELECT

Input: the task description and full module bank. *Output:* the model's selected module descriptions, passed to ADAPT. This selection occurs during plan construction, not separately for each evaluated post. Placeholder spelling, including `reasoning_modules`, is retained from the code.

In order to solve the given task:

<Task>

{Task}

</Task>

Select several modules that are crucial for solving the tasks above
from all the reasoning module description given below:

{reasoning_modules}

C. ADAPT

Input: the task description and SELECT output. *Output:* task-specialized module descriptions, passed to IMPLEMENT. This step asks the model to translate the selected operations into the emotion-classification setting; it does not yet assign a post-level label.

Rephrase and specify each reasoning module so that it better helps solving the task:

<Task>

{Task}

</Task>

SELECTED module descriptions:

{selected_modules}

Adapt each reasoning module description to better solve the task:

D. COMPACT IMPLEMENT

Input: the task description and ADAPT output. *Output:* a reusable reasoning plan, retained for later execution. Here IMPLEMENT means constructing the procedure, not executing it on every post. The prompt requests at most three one-line steps and access to the full post at each step. There is no conformance validator or retry enforcing those requests; the realized plans are reported below.

Operationalize the reasoning modules into a step-by-step reasoning plan in JSON format.

<Task>

{Task}

</Task>

ADAPTED module descriptions:

{adapted_modules}

Implement a reasoning structure that generalises to every instance of this task.

The plan must obey these constraints. They exist because a plan that violates them throws away evidence before it can be used, or is too long to be carried out:

1. ****At most three steps.**** Merge related checks rather than listing them separately.
2. ****Each step is one line.**** A name and a single sentence, nothing more. No nested fields, no examples, no restating of the instructions.
3. Every step takes the full original post as its input. A step may refer to an earlier step's finding, but it must re-read the post rather than work only from an earlier summary.
4. No step before the final one may name, choose, rank or imply the answer.
5. The final step is the only one that weighs the gathered evidence and selects a single label.

Return only the JSON plan.

E. PER-POST EXECUTION

Input: one routed original post, the shared policy, the model's saved plan, and the fixed execution instructions below. *Output:* a newly generated response ending in a classification line. The system message precedes the user message; `structure` is replaced by the saved plan and `text` by this post. No Phase 1 label is supplied to the model.

The evidence ledger is a three-way comparison: for each label the model is asked to identify the strongest supporting and opposing quotation. The subsequent counterfactual check evaluates a competing interpretation rather than changing the post. These are explicit instructions attached to every execution call, not properties guaranteed by caching or by the discovered plan alone.

You are an expert annotator for research-oriented, non-clinical emotion classification. Do not provide clinical diagnosis, medical inference, treatment advice, or professional mental health advice.

Annotate one social-media post for a research-oriented, non-clinical emotion study.

{policy}

Reasoning structure to follow:

{structure}

Post:

<post>

{text}

</post>

Work through the reasoning structure on this post. For every step, quote the exact words from the post that your finding rests on. Do not decide the label before the final step.

Time frame and whose state it is. Before weighing anything, establish two things and quote the evidence for each:

- Is the emotional state the writer reports about their present, or about a past they are describing from a changed position?
- Is it the writer's own state, or a state they attribute to someone else, to a quotation, or to a hypothetical?

These cut in every direction. A post can describe severe hardship and still be Happy when the writer presents it as past and resolved. A post can be light or matter-of-fact in tone and still be Depression when the distress is present and unresolved. A post can recount someone else's crisis while the writer is composed, which is Neutral.

Evidence ledger. Search just as hard for each of the three labels; do not give a shorter treatment to the ones you find less likely.

Depression - strongest supporting quote: "..." | strongest quote against: "..."

Neutral - strongest supporting quote: "..." | strongest quote against: "..."

Happy - strongest supporting quote: "..." | strongest quote against: "..."

Counterfactual check. Name your second-best label. State what would have to appear in this post for that label to be correct instead, then say whether the post contains it.

Then end with exactly one line, in this form:

Final label: Depression

or Final label: Neutral

or Final label: Happy

F. ILLUSTRATIVE PASSAGE THROUGH THE EXECUTION STAGE

Consider the constructed sentence, “Last year I felt isolated; now I enjoy meeting my friends again and feel relieved.” This is an explanatory example, not a dataset record, model response, or additional evaluation result. The same cached plan would be inserted regardless of this sentence’s eventual label. A policy-consistent reading would separate past isolation from current relief, use the quoted time markers as evidence, and examine whether unresolved current distress is actually stated. Under the stated policy, the resolved positive trajectory supports Happy. The example illustrates what the instructions request; it does not show that every generated response follows them.

The evaluator then reads the terminal label from the actual response. A permitted label replaces the Phase 1 prediction for this routed post. If no label is parsed, the evaluator retains Phase 1 instead. Unrouted posts never enter this execution stage. The scored prediction is therefore distinct from the generated explanation, and a parsing failure is not automatically a wrong classification.

G. REALIZED CACHED PLANS

The following plans are reproduced from the recorded notebook outputs, not rewritten to make them conform to the instructions. The Mixed Emotion runs explicitly reused the cached plans. Llama 3 produced three evidence-oriented steps plus a separate final step. Llama 2 produced four prose steps and included provisional label assignment inside early steps. These differences show why the requested plan constraints should not be reported as enforced behavior; they do not independently establish the cause of the performance difference.

Llama 2 generated plan

Here is a step-by-step reasoning plan in JSON format to annotate the emotional meaning of English social-media posts:

Step 1: Separate the writer’s own emotions from those of others (input: full post)

- * Identify emotional language used in the post (e.g. "I feel", "they feel", "it seems")
- * Determine whether the language refers to the writer’s own emotions or someone else’s
- * Assign a label based on the identified emotional language (e.g. "Depression" if the language refers to the writer’s own emotions, "Neutral" if it refers to someone else’s emotions)

Step 2: Locate the emotional stance the post opens with and the emotional stance it closes with (input: full post)

- * Identify the opening and closing sentences of the post
- * Determine whether the emotional stance of the opening sentence differs from the emotional stance of the closing sentence
- * Assign a label based on the identified emotional stance (e.g. "Depression" if the stance differs, "Neutral" if the stance is the same)

Step 3: Determine the final emotional takeaway the writer leaves the reader with (input: full post)

- * Identify the concluding sentence of the post
- * Determine whether the concluding sentence reflects a positive, negative, or neutral emotional state
- * Assign a label based on the identified emotional takeaway (e.g. "Happy" if the takeaway is positive, "Neutral" if it is neutral, "Depression" if it is negative)

Step 4 (final step): Weigh the gathered evidence and select a label (input: full post)

- * Review the labels assigned in steps 1-3
- * Select the label that best reflects the dominant emotional meaning of the post
- * Output the selected label as the research annotation label.

Note: These steps are designed to be carried out in sequence, with each step building on the previous one to gather more information and refine the annotation. The final step is the only one that assigns a label to the post, based on the gathered evidence.

Llama 3 generated plan

Here is the operationalized reasoning plan in JSON format:

```
""
{
  "steps": [
    {
      "name": "Emotion Extraction",
      "sentence": "Extract the writer's explicit and implicit emotional expressions, distinguishing between their own emotions and those attributed to others, quoted text, or hypothetical situations."
    },
    {
      "name": "Emotional Arc and Essence",
      "sentence": "Analyze the emotional trajectory of the post, identifying the emotional stance at the beginning and end, and determine the dominant emotional takeaway the writer intends to convey."
    },
    {
      "name": "Contextual Clarification and Purpose Identification",
      "sentence": "Disentangle past events or experiences from the writer's current emotional state, identify the primary purpose of the post, and consider the implications for their emotional expression."
    }
  ],
  "final_step": {
    "name": "Label Prioritization",
    "sentence": "Rank the three candidate labels (Depression, Neutral, Happy) in order of likelihood, considering the emotional essence, emotional arc, and contextual factors, and provide a second-best label to ensure the annotation is robust and defensible."
  }
}
""
```

This plan consists of three main steps that extract the writer's emotional expressions, analyze the emotional trajectory and essence, and clarify the context and purpose of the post. The final step prioritizes the candidate labels based on the gathered evidence.

H. GENERATION BUDGET AND FAILURE HANDLING

Direct and CoT request 1,024 new tokens per generation call. Final SD execution requests 1,024 for Llama 2 and 1,536 for Llama 3. The SD discovery calls request up to 512 tokens for Llama 2 and 768 for Llama 3. The source uses a shared per-model cache for the task-level structures, allowing reuse across the Reddit and Mixed Emotion runs. The templates specify instructions, not guaranteed instruction compliance. Quantized model loading, model-specific chat formatting, context limits, and call structure are part of the evaluated implementation. A shorter plan instruction does not by itself prove lower measured cost.

Remaining final SD parsing failures are 21 for Llama 2 Reddit, zero for Llama 2 Mixed Emotion, two for Llama 3 Reddit, and one for Llama 3 Mixed Emotion. There are no SD parsing failures in the additional 9,000-post evaluation. These remain failures and use Phase 1 fallback. No additional model call is added to resolve them in the reported final SD results.

APPENDIX D SELF-DISCOVER DESIGN-VARIANT COMPARISON

The earlier independent-countercheck variant generated an instance-specific plan using general reasoning modules; the final version uses a domain-oriented task-level plan and an explicit evidence ledger. Both withhold the Phase 1 label. Multiple components change jointly, so this is a design-variant comparison rather than a single-component causal ablation. The earlier variant is not used in the main six-configuration comparison.

Appendix Table D1. Earlier and final SELF-DISCOVER designs

Component	Earlier independent-countercheck	Final task-level compact plan
Discovery scope	A plan constructed for each post	A model-specific plan constructed for the task and reused
Module bank	General reasoning operations	18 emotion-oriented operations
Execution guidance	Independent assessment and countercheck	Explicit subject/time checks, three-label quotation ledger, and counterfactual check
Phase 1 label	Withheld during SD reasoning	Withheld during SD reasoning
Evaluation role	Earlier design comparator in this appendix	SELF-DISCOVER condition in the main comparison

Appendix Table D2. Full-set outcomes of the two SD designs

Dataset	Model	Earlier acc.	Final acc.	Earlier net	Final net
Reddit	Llama 2	97.0500%	96.9417%	16	3
Reddit	Llama 3	97.1583%	97.2000%	29	34
Mixed Emotion	Llama 2	87.6667%	89.3333%	12	17
Mixed Emotion	Llama 3	90.3333%	92.6667%	20	27

Table note. Accuracy uses all 12,000 Reddit posts or all 300 Mixed Emotion examples, not only routed posts. Net denotes corrected minus introduced errors relative to the same final Phase 1 predictions. “Earlier” refers only to the independent-countercheck design, not another intermediate compact-plan version. Table D2 reports observed outcomes, not a paired significance test between the two SD designs.

The final design improves three of these four model–dataset combinations; it does not uniformly dominate the earlier design. Its Llama 3 configuration improves both datasets. The change in input-independent planning also changes the number and scope of discovery calls, so a cost comparison requires measured generation records rather than accuracy alone.

APPENDIX E PAIRED ACCURACY COMPARISONS

All rows below use ID-aligned final predictions, including the implemented Phase 1 fallback for parsing failures. Table E1 compares each main configuration with Phase 1; Table E2 instead compares SD with Direct or CoT on the same examples. A positive change favors the named re-evaluator in E1 and SD in E2. Intervals are 50,000-resample paired percentile-bootstrap intervals for accuracy change, in percentage points. Exact p is a two-sided McNemar test. These tests do not measure macro F1 uncertainty or adjust for prompt development on the main evaluation cases.

Appendix Table E1. Main comparisons with Phase 1

Dataset	Model	Protocol	Change (pp)	Paired 95% CI	Exact p	Holm p
Reddit	Llama 2	Direct	+0.0750	[-0.075, 0.225]	0.3742	1.0000
Reddit	Llama 2	CoT	+0.0083	[-0.150, 0.167]	1.0000	1.0000
Reddit	Llama 2	SD	+0.0250	[-0.142, 0.192]	0.8424	1.0000
Reddit	Llama 3	Direct	+0.1917	[0.017, 0.367]	0.0380	0.2281
Reddit	Llama 3	CoT	+0.2333	[0.075, 0.392]	0.0061	0.0479
Reddit	Llama 3	SD	+0.2833	[0.117, 0.458]	0.0015	0.0151
Mixed Emotion	Llama 2	Direct	-7.3333	[-11.667, -3.000]	0.0013	0.0139
Mixed Emotion	Llama 2	CoT	-4.6667	[-9.333, 0.000]	0.0704	0.3520
Mixed Emotion	Llama 2	SD	+5.6667	[2.333, 9.000]	0.0015	0.0151
Mixed Emotion	Llama 3	Direct	+5.6667	[2.000, 9.667]	0.0060	0.0479
Mixed Emotion	Llama 3	CoT	+1.0000	[-2.333, 4.333]	0.7011	1.0000
Mixed Emotion	Llama 3	SD	+9.0000	[5.667, 12.333]	< 0.0001	< 0.0001

Holm adjustment in Table E1 covers all 12 rows.

Appendix Table E2. SD versus alternative re-evaluators

Dataset	Model	Comparison	Change (pp)	Paired 95% CI	Exact p	Holm p
Reddit	Llama 2	SD – Direct	-0.0500	[-0.192, 0.092]	0.5560	1.0000
Reddit	Llama 2	SD – CoT	+0.0167	[-0.142, 0.175]	0.9179	1.0000
Reddit	Llama 3	SD – Direct	+0.0917	[-0.008, 0.192]	0.1081	0.2163
Reddit	Llama 3	SD – CoT	+0.0500	[-0.033, 0.142]	0.3616	0.3616
Mixed Emotion	Llama 2	SD – Direct	+13.0000	[9.333, 17.000]	< 0.0001	< 0.0001
Mixed Emotion	Llama 2	SD – CoT	+10.3333	[6.000, 14.667]	< 0.0001	< 0.0001
Mixed Emotion	Llama 3	SD – Direct	+3.3333	[1.333, 5.333]	0.0020	0.0020
Mixed Emotion	Llama 3	SD – CoT	+8.0000	[5.000, 11.333]	< 0.0001	< 0.0001
Additional 9,000	Llama 3	SD – Direct	+0.0444	[-0.056, 0.144]	0.5235	0.9614
Additional 9,000	Llama 3	SD – CoT	+0.0444	[-0.044, 0.133]	0.4807	0.9614

Holm adjustment in Table E2 is applied to the two comparisons within each dataset–model combination, not to all ten rows as one family.

Appendix Table E3. Additional 9,000 posts versus Phase 1

Protocol	Change (pp)	Paired 95% CI	Exact p	Holm p
Direct	+0.1444	[-0.0111, 0.3000]	0.0919	0.1320
CoT	+0.1444	[0.0000, 0.2889]	0.0660	0.1320
SD	+0.1889	[0.0444, 0.3333]	0.0137	0.0412

Table E3 treats its three comparisons as a separate Holm family. Non-significance of an SD-versus-alternative comparison is not evidence of equivalence.

APPENDIX F SYNTHETIC MIXED EMOTION DATASET GENERATION PROTOCOL

Appendix Table F1. Synthetic Mixed Emotion Dataset generation protocol

Component	Description
Purpose	Construct 300 controlled examples spanning mixed or shifting affect and Neutral controls for controlled stress-test evaluation.
Use in pipeline	Not used for Phase 1 training, hyperparameter tuning, or routing-threshold selection; used only for controlled stress-test evaluation.
Class balance	Depression = 100, Neutral = 100, Happy = 100.
Scenario balance	Seven controlled scenario groups: five 40-example mixed or shifting-emotion groups, a 75-example clear technical/informational Neutral group, and a 25-example mild factual/procedural-ambiguity Neutral group.
Scenario types	Blended-emotion co-occurrence; positive-to-distress shift; distress-to-recovery shift; Neutral framing with subtle affect; conflicting cues with a dominant trajectory; clear technical or informational Neutral content; and mild factual or procedural Neutral ambiguity.
Label rule	Assign Depression, Neutral, or Happy according to the dominant overall emotional trajectory, not isolated phrases.
Exclusion criteria	Exclude explicit diagnosis claims, treatment advice, medication references, crisis language, identifying information, and off-scenario content. Mixed-cue requirements do not apply to the clear Neutral controls.
Prompt disclosure	Generation instructions are reproduced below; the version, model, date, and inclusion criteria are recorded here.
Generation record	GPT-5; prompt version <code>mixed-emotion-stress-test-v2.4-neutral-clear</code> ; final records dated Aug. 7, 2026.
Limitations	Synthetic origin, generation-model style bias, and lack of expert annotation.

Generation prompt used for synthetic stress-test examples.

Generate synthetic Reddit-style posts for a controlled mixed-emotion stress-test dataset. Use one of three target labels: Depression, Neutral, or Happy. For Depression and Happy examples, include mixed or shifting cues but make the final emotional trajectory clear to a trained reviewer. For Neutral examples, retain low affective intensity: include either technical/informational content or mild factual/procedural ambiguity without sustained distress, relief, accomplishment, or other dominant affect. When multiple emotions occur, the target label must follow the overall message and final takeaway rather than an isolated phrase.

For each example, write one realistic post between 60 and 90 words. Do not include explicit diagnosis claims, medication names, therapy claims, suicide, self-harm, crisis language, usernames, URLs, subreddit names, hashtags, or personally identifying information. Avoid duplicated phrasing and avoid making the label obvious through a single keyword. Return structured records with: example identifier, target label, scenario type, primary context, dominant emotional trajectory, text, and brief label rationale.

APPENDIX G CLASS-LEVEL RESULTS AND ADDITIONAL EVALUATION

Table G1 reports class-wise F1 for each complete pipeline. All metrics use the full dataset, not only routed posts. Failures count routed responses without a parsed label; those records retain their Phase 1 predictions and remain in every metric. Reddit and Mixed Emotion contain 4,000 and 100 reference examples per class, respectively; the additional set contains 3,000 per class.

Appendix Table G1. Class-wise F1 and retained parsing failures

Dataset	Model	Protocol	Depression F1	Neutral F1	Happy F1	Failures
Reddit	Llama 2	Direct	96.40	98.53	96.05	1
Reddit	Llama 2	CoT	96.24	98.55	95.98	0
Reddit	Llama 2	SD	96.47	98.37	95.97	21
Reddit	Llama 3	Direct	96.60	98.51	96.21	1
Reddit	Llama 3	CoT	96.64	98.49	96.31	1
Reddit	Llama 3	SD	96.66	98.67	96.27	2
Mixed Emotion	Llama 2	Direct	84.10	82.91	58.48	0
Mixed Emotion	Llama 2	CoT	75.65	88.24	72.91	0
Mixed Emotion	Llama 2	SD	87.23	96.04	84.76	0
Mixed Emotion	Llama 3	Direct	90.11	93.27	84.76	0
Mixed Emotion	Llama 3	CoT	90.11	87.39	76.53	0
Mixed Emotion	Llama 3	SD	90.11	97.98	90.00	1
Additional 9,000	Llama 3	Direct	97.34	98.78	97.05	2
Additional 9,000	Llama 3	CoT	97.34	98.73	97.10	0
Additional 9,000	Llama 3	SD	97.34	98.86	97.10	0

A. FIXED ADDITIONAL-EVALUATION PROTOCOL

The additional run loads the same final seed-42 DistilBERT weights and uses $T^* = 1.480195$, $\tau^* = 0.70$, a 256-token limit, and fp32 inference with batch size 64. Phase 1 training, calibration fitting, and threshold search are disabled. The Llama 3 protocols use the exact templates in Appendices B–C and the retained model-specific SD plan, with no new discovery calls. All three methods receive the same 111 routed original texts; the 6,000-character input policy, greedy decoding, 4-bit NF4 loading, and per-call limits (1,024 for Direct/CoT; 1,536 for SD) remain fixed. Execution uses an NVIDIA L4 GPU. Runtime, model revision, input hashes, and the saved plan are recorded with the run.

There are 111 Direct calls, 555 CoT calls, and 111 SD calls. These counts include the five-turn CoT dialogue but no new SD discovery. All 777 saved calls were checked against their request hashes and run configuration; final outputs were aligned by ID and recombined with accepted Phase 1 predictions. The separate example-level tests in Appendix E use these complete predictions. Call counts alone do not establish a matched time or monetary-cost advantage.

(a) Direct				(b) CoT				(c) SELF-DISCOVER						
		Predicted label					Predicted label					Predicted label		
		Depression	Neutral	Happy			Depression	Neutral	Happy			Depression	Neutral	Happy
Reference label	Depression	2,894 96.5%	13 0.4%	93 3.1%	Reference label	Depression	2,889 96.3%	18 0.6%	93 3.1%	Reference label	Depression	2,890 96.3%	13 0.4%	97 3.2%
	Neutral	13 0.4%	2,958 98.6%	29 1.0%		Neutral	11 0.4%	2,960 98.7%	29 1.0%		Neutral	12 0.4%	2,959 98.6%	29 1.0%
	Happy	39 1.3%	18 0.6%	2,943 98.1%		Happy	36 1.2%	18 0.6%	2,946 98.2%		Happy	36 1.2%	14 0.5%	2,950 98.3%

Appendix Figure G1. End-to-end confusion matrices for Direct, CoT, and final SELF-DISCOVER on the same 9,000 posts. Rows are reference labels; columns are final predictions. Counts and within-class percentages use a common color scale.

...